Iterated Symmetric Centipede Game: Full Evolutionary Analysis

Four Variations: All-$M/2$ Modification, Rigorous Three-Strategy Analysis,
Five-Strategy Extension, and Two-Population Framework

2026

Abstract

This report presents a unified and complete evolutionary game-theoretic analysis of the Iterated Symmetric Centipede (ISC) game across four distinct project variations, all grounded in the framework of Lazaridis and Kehagias [1]. The ISC involves T repeated plays of the One-Shot Symmetric Centipede (OSC), in which the first-moving player is selected uniformly at random.

Variation 1 replaces the All-M strategy of the original paper with All-$M/2$, studying co-evolutionary dynamics with All-1 and Grim under both Replicator Dynamics and Markov Chain Dynamics. **Variation 2** provides a rigorous mathematical treatment of the same three-strategy game, deriving all Nash equilibria, stability conditions, and evolutionarily stable strategies with full proofs. **Variation 3** extends the strategy set to five strategies— All-M, All-1, Grim, Grim-Patient, and Grim-Forgiver—exploiting the rich degeneracy structure of the resulting payoff matrix. **Variation 4** replaces the single evolving population with two populations playing exclusively against each other, uncovering asymmetric Nash equilibria and richer absorbing structures.

Across all variations, a critical threshold $p^* = 2/3$ governs the stability of the all-defector equilibrium. The Grim strategy (and its variants) emerges as dominant in cooperative regimes, while the two-population framework and the five-strategy extension reveal new equilibrium structures not present in the original one-population three-strategy game.

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1 Introduction

This report consolidates four related evolutionary analyses of the Iterated Symmetric Centipede (ISC) game, based on the framework of Lazaridis and Kehagias [1]. Each variation modifies a different structural aspect of the original model — the strategy set, the population structure, or the mathematical depth of analysis — and together they form a comprehensive study of how cooperation emerges and is sustained in repeated strategic interactions.

The One-Shot Symmetric Centipede (OSC) is a two-player extensive-form game with M stages. At each stage i, the active player may *terminate* (taking action ρ_i, ending the game) or *pass* the move to the opponent. The first-moving player is selected uniformly at random, so each player moves first with probability $1/2$. The terminator receives share $p \in (1/2, 1]$ of the joint payoff, while the passive player receives $1 - p$. The resulting expected payoff matrix A to the row player is:

$$A(i,j) = \begin{cases} i & \text{if } i = j, \\ (2i+1)\,p & \text{if } i < j, \\ (2j+1)\,(1-p) & \text{if } i > j. \end{cases}$$

Four values from this matrix recur throughout the report:

$$A_{MM} = M, \quad A_{11} = 1, \quad A_{M1} = 3(1-p), \quad A_{1M} = 3p.$$

The ISC consists of T repeated plays of the OSC, with full history observable between rounds. The unique subgame-perfect equilibrium of the OSC is (ρ_1, ρ_1), yielding $(p, 1-p)$, which is Pareto-dominated by (ρ_M, ρ_M) yielding $(pM, (1-p)M)$. The central question of the ISC is whether iterated play and evolutionary pressure can sustain cooperative outcomes.

We study evolutionary dynamics under two frameworks: (i) *Replicator Dynamics* (RD), governing large-population continuous evolution, and (ii) *Markov Chain Dynamics* with Pairwise Proportional Imitation (PPI), governing small finite-population stochastic evolution.

The report is organised as follows. Sections 2–5 each present one project variation in full, including the relevant payoff matrix, Nash equilibria, replicator dynamics, evolutionary stability, and Markov chain analysis. Section 6 presents cross-variation observations and comparative remarks. Section 7 states the overall conclusions.

2 Variation 1: All-$M/2$, All-1, and Grim — Numerical Analysis

2.1 Overview and Modification

This variation replaces the **All-M** strategy of the original paper [1] with **All-$M/2$**, which cooperates at the midpoint level $\rho_{M/2}$ rather than the maximum ρ_M. Three ISC strategies are studied (default parameters: $M = 4$, $k = M/2 = 2$):

Index	Symbol	Name	Description
1	$\sigma_{\text{All-}M/2}$	All-$M/2$	Always play $\rho_{M/2}$: cooperates at the mid-level.
2	$\sigma_{\text{All-1}}$	All-1	Always play ρ_1: always defects immediately.
3	σ_G	Grim	Play ρ_M in round 1; if the opponent ever plays ρ_m with $m < M$, switch permanently to ρ_1.

Grim trigger rule. The trigger condition is $m < M$:

- Grim *triggers* against All-$M/2$ (plays $\rho_{M/2}$, and $M/2 < M$).
- Grim *triggers* against All-1 (plays ρ_1, and $1 < M$).
- Grim does *not* trigger against another Grim (plays ρ_M, and $M \not< M$).

Consequently, two Grim players cooperate at the highest level every round, yielding the self-play payoff $C(3,3) = T \cdot M$.

2.2 The ISC Payoff Matrix

The entry $C(i,j)$ is the payoff to strategy i when facing strategy j over T rounds. The symbolic matrix is:

$$C = \begin{bmatrix} \frac{M}{2}\,T & 3(1-p)\,T & (M+1)p + (T-1)\,3(1-p) \\ 3p\,T & T & 3p + (T-1) \\ (M+1)(1-p) + (T-1)\,3p & 3(1-p) + (T-1) & T\,M \end{bmatrix}$$

Table 1: Round-by-round derivation of each $C(i,j)$ entry (Variation 1).

Cell	Matchup	Round 1	Trigger?	Rounds $2 \ldots T$	Formula
$C(1,1)$	All-$M/2$ vs All-$M/2$	$A(k,k)$	No	$A(k,k)$	$T \cdot (M/2)$
$C(1,2)$	All-$M/2$ vs All-1	$A(k,1)$	No	$A(k,1)$	$3(1-p)\,T$
$C(1,3)$	All-$M/2$ vs Grim	$A(k,M)$	Yes	$A(k,1)$	$(M+1)p + (T-1) \cdot 3(1-p)$
$C(2,1)$	All-1 vs All-$M/2$	$A(1,k)$	No	$A(1,k)$	$3pT$
$C(2,2)$	All-1 vs All-1	$A(1,1)$	No	$A(1,1)$	T
$C(2,3)$	All-1 vs Grim	$A(1,k)$	Yes	$A(1,1)$	$3p + (T-1)$
$C(3,1)$	Grim vs All-$M/2$	$A(M,k)$	Yes	$A(1,k)$	$(M+1)(1-p) + (T-1) \cdot 3p$
$C(3,2)$	Grim vs All-1	$A(M,1)$	Yes	$A(1,1)$	$3(1-p) + (T-1)$
$C(3,3)$	Grim vs Grim	$A(M,M)$	No	$A(M,M)$	$T \cdot M$

Table 2: Numerical payoff matrix C for $p = 3/4$, $M = 4$, $T = 10$.

	vs All-$M/2$	vs All-1	vs Grim
All-$M/2$	20	7.5	10.5
All-1	22.5	10	11.25
Grim	21.5	9.75	40

Table 3: Numerical payoff matrix C for $p = 3/5$, $M = 4$, $T = 10$.

	vs All-$M/2$	vs All-1	vs Grim
All-$M/2$	20	12	13.8
All-1	18	10	10.8
Grim	18.2	10.2	40

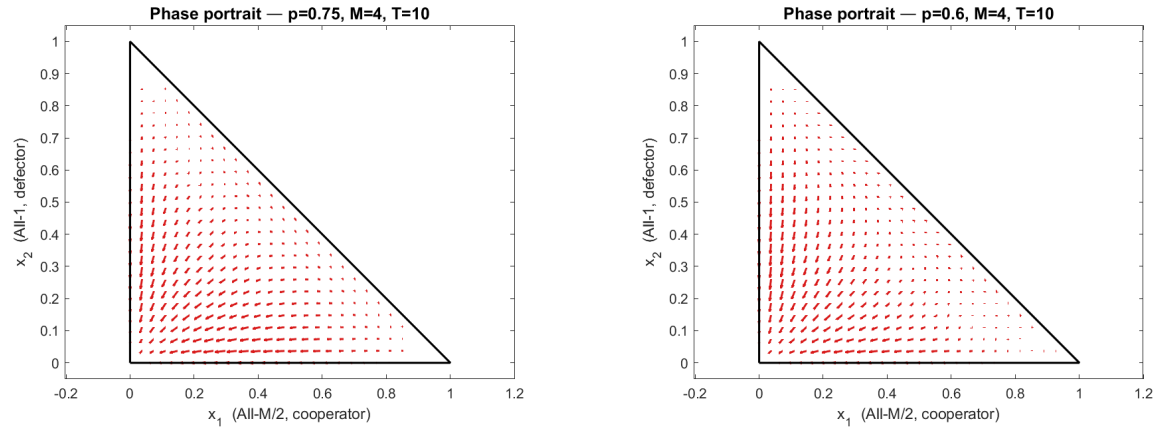
Why Grim dominates. The diagonal self-play payoffs are: All-$M/2$: $T \cdot (M/2) = 20$; All-1: $T = 10$; Grim: $T \cdot M = 40$. Grim earns twice the payoff of All-$M/2$ in a homogeneous population because it cooperates at ρ_M without triggering against other Grims.

Contrast with original paper. In [1], All-M and Grim both earn $TM = 40$ in self-play, so the $s_2 = 0$ face is absorbing. Here, $C(1,1) = 20 \neq 40 = C(3,3)$: All-$M/2$ players will switch to Grim on that face, so the only absorbing states are the three pure vertices $(N, 0, 0)$, $(0, N, 0)$, $(0, 0, N)$.

2.3 Replicator Dynamics

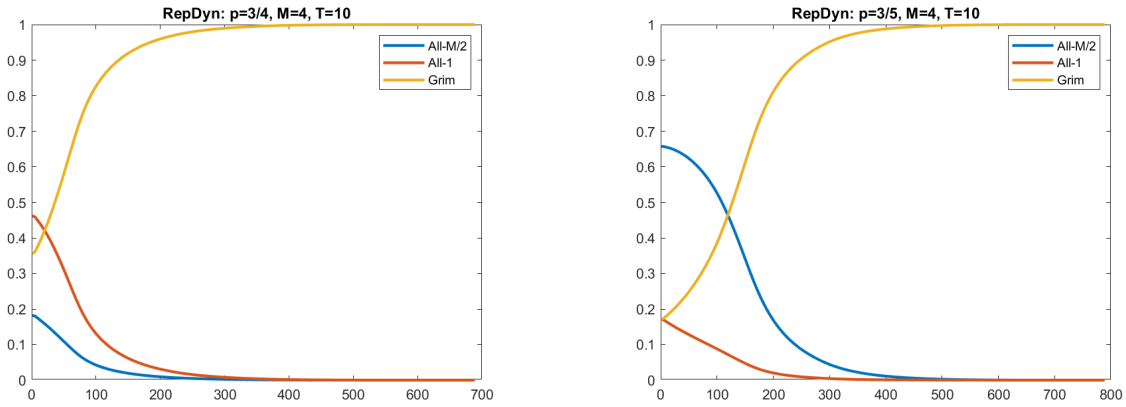
Let $x = (x_1, x_2, x_3)$ denote the population frequencies of All-$M/2$, All-1, and Grim respectively, with $x_1 + x_2 + x_3 = 1$ and $x_i \geq 0$. The replicator equation is:

$$\dot{x}_m = x_m \left[(Cx)_m - x^\top C x \right], \quad m = 1, 2, 3.$$



(a) $p = 3/4$ (b) $p = 3/5$

Figure 1: Phase portraits of the replicator dynamics, Variation 1 ($M = 4$, $T = 10$). Axes: x_1 = All-$M/2$ frequency, x_2 = All-1 frequency. The Grim frequency is $x_3 = 1 - x_1 - x_2$.



(a) $p = 3/4$ (b) $p = 3/5$

Figure 2: Replicator dynamics time series, Variation 1 ($M = 4$, $T = 10$).

Parameter Sensitivity

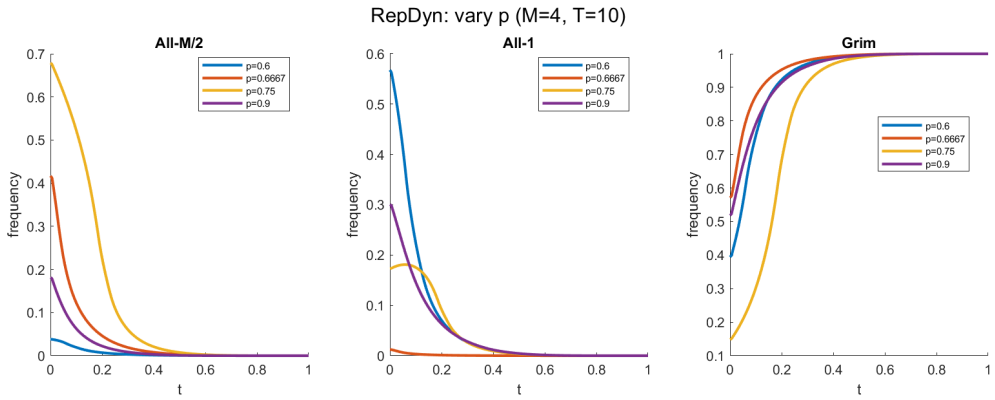


Figure 3: Replicator dynamics: effect of varying $p \in \{3/5, 2/3, 3/4, 9/10\}$ ($M = 4$, $T = 10$), Variation 1.

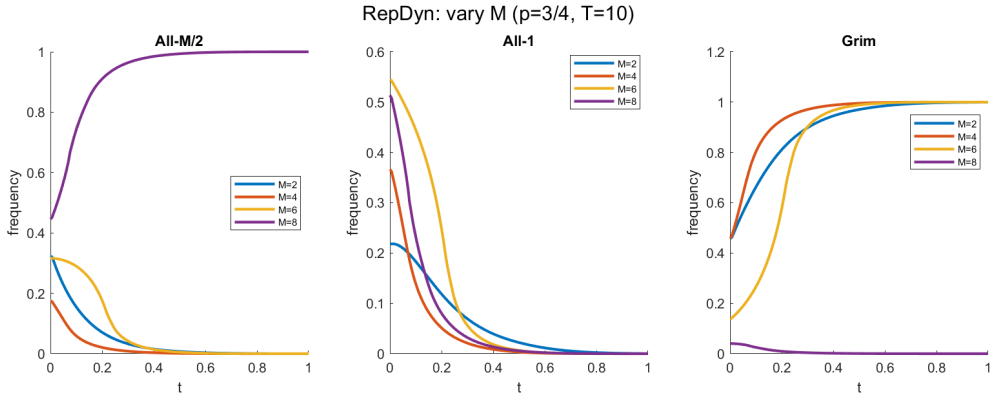


Figure 4: Replicator dynamics: effect of varying $M \in \{2, 4, 6, 8\}$ ($p = 3/4$, $T = 10$), Variation 1.

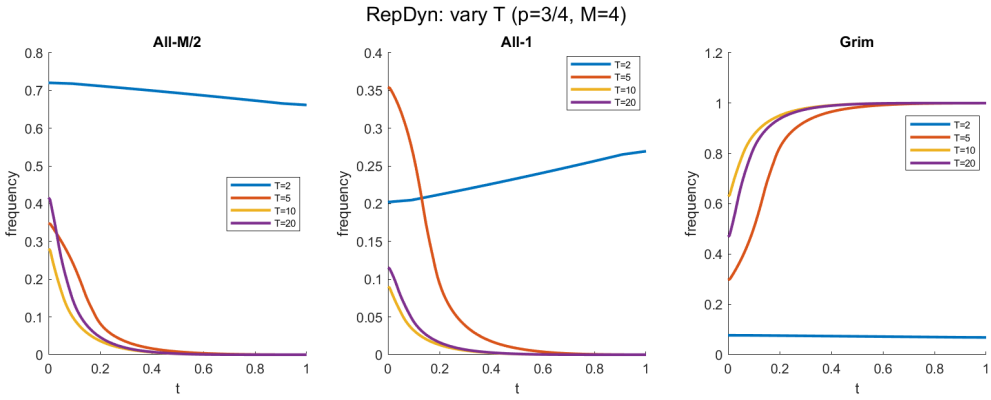


Figure 5: Replicator dynamics: effect of varying $T \in \{2, 5, 10, 20\}$ ($p = 3/4$, $M = 4$), Variation 1.

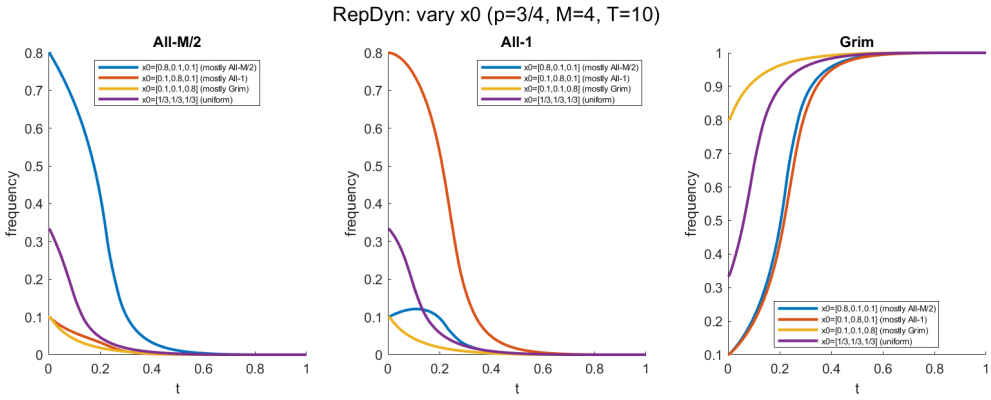


Figure 6: Replicator dynamics: effect of varying initial condition x_0 ($p = 3/4$, $M = 4$, $T = 10$), Variation 1.

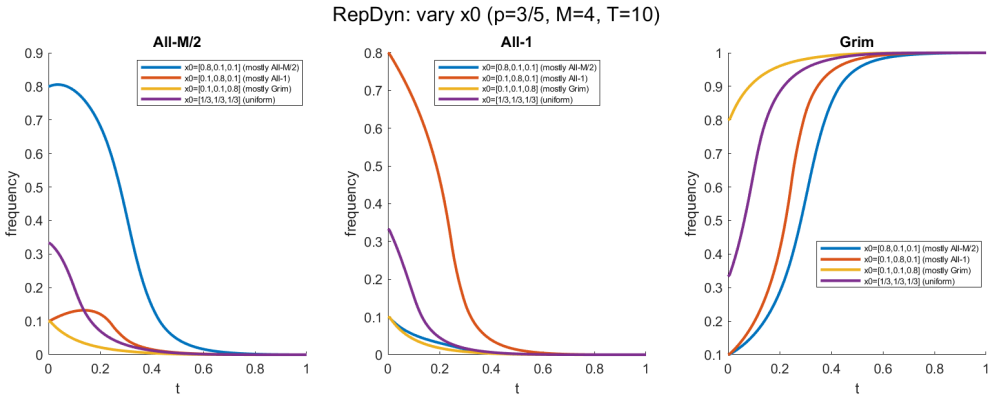


Figure 7: Replicator dynamics: effect of varying initial condition x_0 ($p = 3/5$, $M = 4$, $T = 10$), Variation 1.

2.4 Markov Chain Dynamics

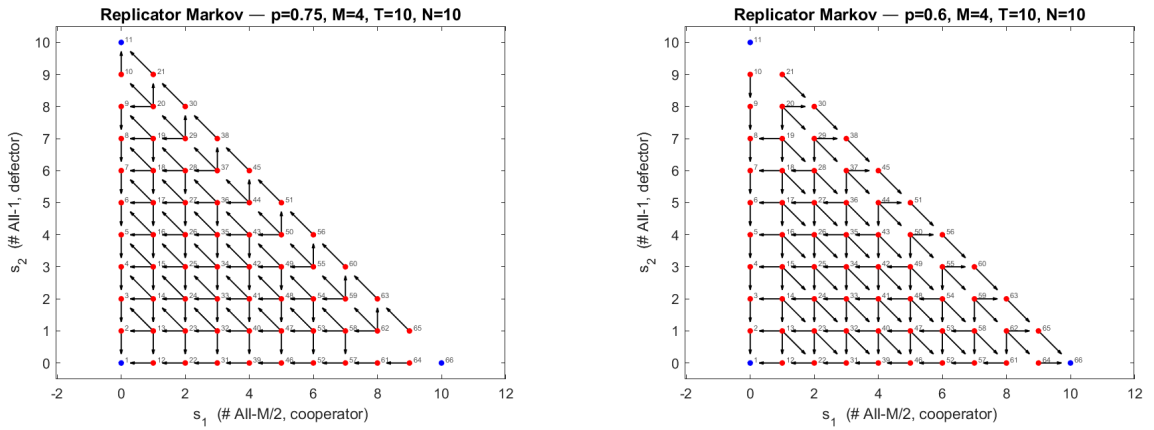
The population state is $s = (s_1, s_2, s_3)$ with $s_1 + s_2 + s_3 = N$ and $s_i \geq 0$ integers. The per-player payoff of strategy m in state s is:

$$q_m(s) = (s_m - 1)\, C(m, m) + \sum_{n \neq m} s_n\, C(m, n).$$

Under the **Pairwise Proportional Imitation (PPI)** revision protocol, a randomly selected player using strategy m_1 switches to m_2 with probability:

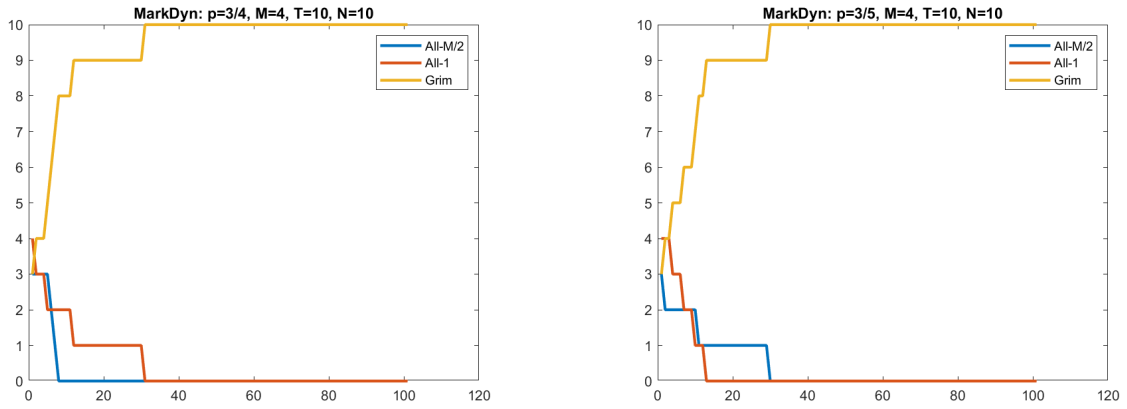
$$\varrho_{m_1 m_2}(s) = \frac{x_{m_2}\,[q_{m_2}(s) - q_{m_1}(s)]_+}{\sum_m x_m\,[q_m(s) - q_{m_1}(s)]_+}.$$

Absorbing states. Because $C(1,1) = T \cdot (M/2) = 20 \neq 40 = C(3,3)$, Grim earns strictly more than All-$M/2$ on the $s_2 = 0$ face. The **only absorbing states** are the three pure vertices: $(N, 0, 0)$, $(0, N, 0)$, $(0, 0, N)$.



(a) $p = 3/4$ (b) $p = 3/5$

Figure 8: Markov state transition graphs, Variation 1 ($M = 4$, $T = 10$, $N = 10$). Blue dots: absorbing (pure) states. Red dots: transient states. Arrow thickness indicates transition probability.



(a) $p = 3/4$ (b) $p = 3/5$

Figure 9: Markov dynamics time series, Variation 1 ($M = 4$, $T = 10$, $N = 10$, $T_m = 100$, $s_0 = [3, 4, 3]$).

Parameter Sensitivity

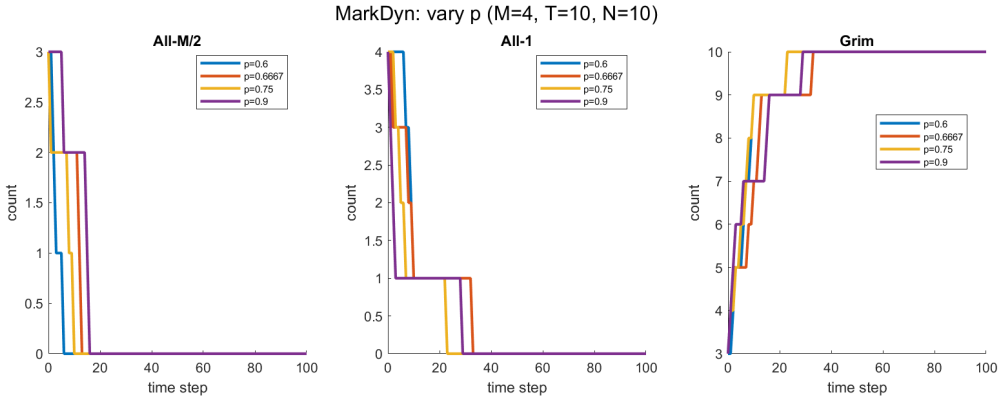


Figure 10: Markov dynamics: effect of varying $p \in \{3/5,\ 2/3,\ 3/4,\ 9/10\}$ ($M = 4$, $T = 10$, $N = 10$), Variation 1.

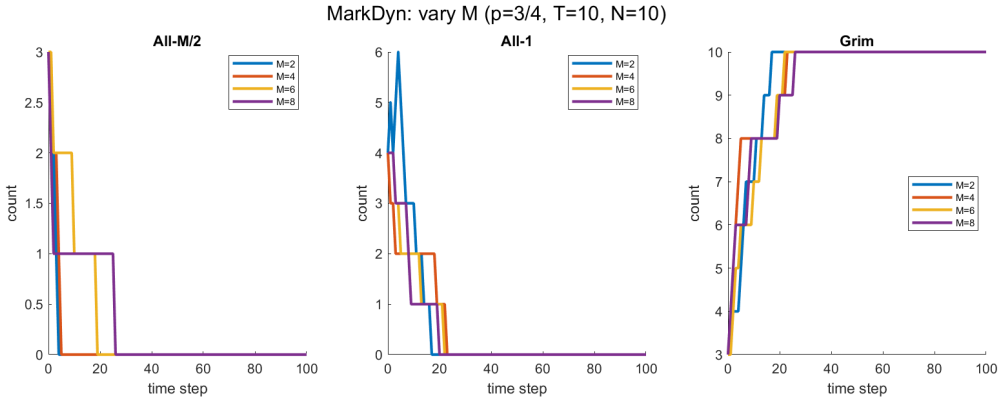


Figure 11: Markov dynamics: effect of varying $M \in \{2, 4, 6, 8\}$ ($p = 3/4$, $T = 10$, $N = 10$), Variation 1.

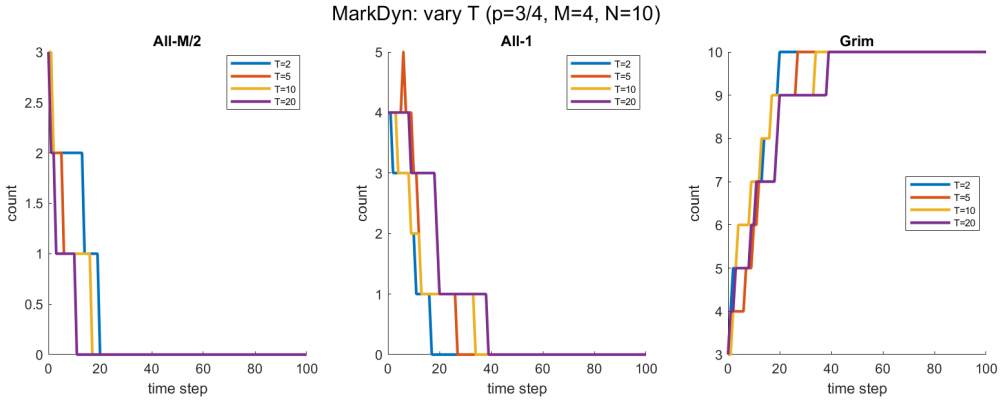


Figure 12: Markov dynamics: effect of varying $T \in \{2, 5, 10, 20\}$ ($p = 3/4$, $M = 4$, $N = 10$), Variation 1.

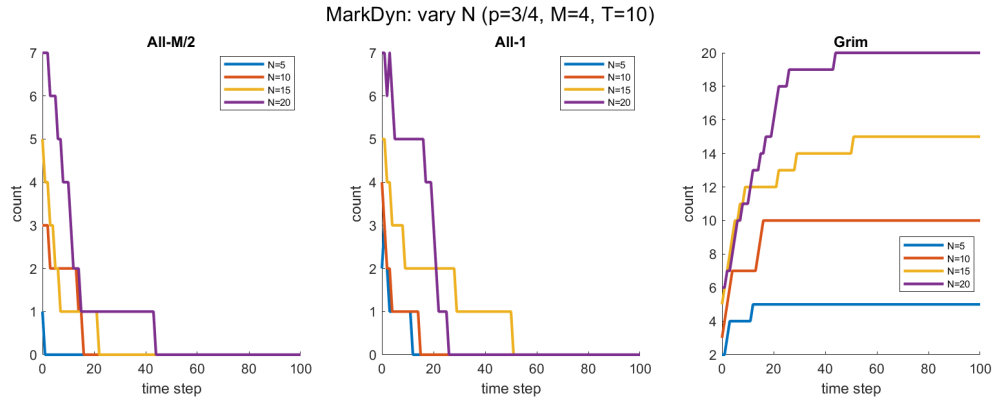


Figure 13: Markov dynamics: effect of varying $N \in \{5, 10, 15, 20\}$ ($p = 3/4$, $M = 4$, $T = 10$), Variation 1.

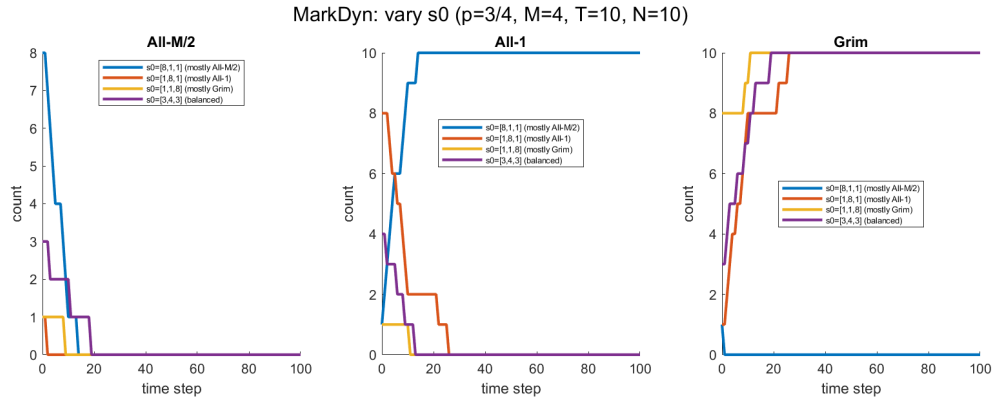


Figure 14: Markov dynamics: effect of varying initial state s_0 ($p = 3/4$, $M = 4$, $T = 10$, $N = 10$), Variation 1.

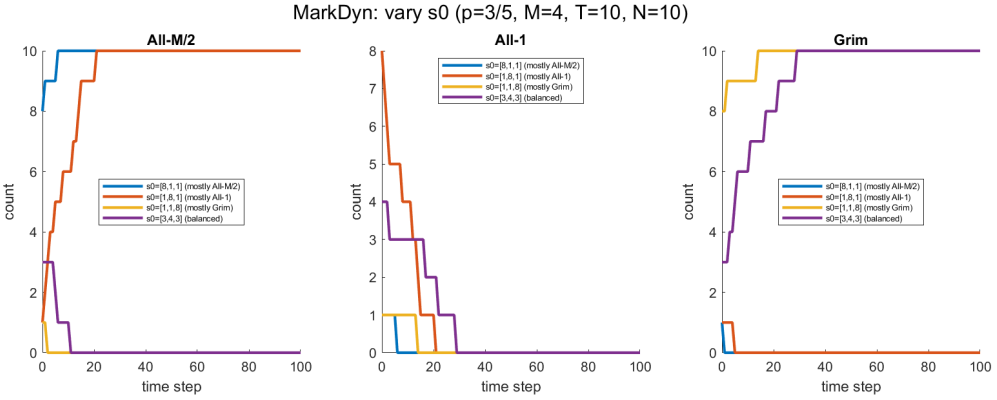


Figure 15: Markov dynamics: effect of varying initial state s_0 ($p = 3/5$, $M = 4$, $T = 10$, $N = 10$), Variation 1.

2.5 Key Findings of Variation 1

Table 4: Comparison: original paper (All-M) vs. Variation 1 (All-$M/2$).

Aspect	Original (All-M)	Variation 1 (All-$M/2$)
Cooperator strategy	Always play ρ_M	Always play $\rho_{M/2}$
$C(1,1)$	$T \cdot M = 40$	$T \cdot (M/2) = 20$
$C(3,3)$	$T \cdot M = 40$	$T \cdot M = 40$ (unchanged)
Grim trigger vs cooperator	Never ($M \not< M$)	Always ($M/2 < M$)
s_2=0 face absorbing?	Yes (entire face)	No (only pure vertices)
Absorbing states	$(0, N, 0)$ + s_2=0 face	$(N, 0, 0)$, $(0, N, 0)$, $(0, 0, N)$

A critical threshold emerges at $p = 2/3$:

- $p > 2/3$ (e.g. $p = 3/4$): the pure-defector state $(0, 1, 0)$ is asymptotically stable. Both Grim and All-1 can be long-run outcomes.
- $p < 2/3$ (e.g. $p = 3/5$): $(0, 1, 0)$ is unstable. All-1 is driven out; the population converges to the All-$M/2$–Grim edge.

3 Variation 2: All-$M/2$, All-1, and Grim — Rigorous Mathematical Analysis

3.1 Overview

This variation provides a complete rigorous mathematical treatment of the same three-strategy ISC game as Variation 1, deriving all Nash equilibria, stability conditions of rest points, and evolutionarily stable strategies with full proofs. The strategy set and payoff matrix C are identical to Variation 1 (see Tables 2–3); the large-T approximation is:

$$C \approx T \begin{bmatrix} k & 3(1-p) & 3(1-p) \\ 3p & 1 & 1 \\ 3p & 1 & M \end{bmatrix}, \quad k = M/2. \tag{1}$$

$$C = \begin{bmatrix} \frac{M}{2}\,T & 3(1-p)\,T & (M+1)p + (T-1)\,3(1-p) \\ 3p\,T & T & 3p + (T-1) \\ (M+1)(1-p) + (T-1)\,3p & 3(1-p) + (T-1) & T\,M \end{bmatrix}$$

Figure 16: Symbolic payoff matrix C for Variation 2.

3.2 Nash Equilibria

Definition 3.1 (Nash Equilibrium)**.** *A pair of mixed strategies* $(\hat{\sigma}_1, \hat{\sigma}_2)$ *is a* Nash equilibrium *(NE) iff*

$$\forall \sigma_1 :\ Q_1(\hat{\sigma}_1, \hat{\sigma}_2) \geq Q_1(\sigma_1, \hat{\sigma}_2) \quad \textit{and} \quad \forall \sigma_2 :\ Q_2(\hat{\sigma}_1, \hat{\sigma}_2) \geq Q_2(\hat{\sigma}_1, \sigma_2).$$

Proposition 3.1 (Pure-Strategy NE)**.** *For $M \geq 4$, $k = M/2$, and T large enough:*

1. $\big((0,0,1),(0,0,1)\big)$ *is a NE for all* $p \in (\frac{1}{2}, 1]$.
2. $\big((0,1,0),(0,1,0)\big)$ *is a NE if and only if* $p \geq \frac{2}{3}$.
3. $\big((1,0,0),(1,0,0)\big)$ *is a NE if and only if* $p \leq \frac{k}{3} = \frac{M}{6}$.

Proof. (1) Pure Grim. For a deviation $(a, b, 1-a-b)$: $\delta Q_1 = a(TM - C(1,3)) + b(TM - C(2,3)) \geq 0$, since $TM - C(1,3) \approx T(M-3+3p) > 0$ and $TM - C(2,3) \approx T(M-1) > 0$ for $M \geq 2$.

*(2) Pure All-*1. Deviation toward strategy 1: $\delta Q_1 = aT(3p-2)$. Deviation toward strategy 3: $\delta Q_1 = c(2-3p)$. Both are non-negative iff $p \geq 2/3$.

(3) Pure All-k. Deviation toward strategies 2 or 3: $\delta Q_1 \approx T(k-3p)$, which is non-negative iff $p \leq k/3 = M/6$. □

Remark 3.1. *For $M = 4$ ($k = 2$), the thresholds for parts (2) and (3) coincide at $p^* = 2/3$:*

- *$p < 2/3$: $(0,0,1)$ and $(1,0,0)$ are pure NE; $(0,1,0)$ is not.*
- *$p > 2/3$: $(0,0,1)$ and $(0,1,0)$ are pure NE; $(1,0,0)$ is not.*
- *$p = 2/3$: only $(0,0,1)$ is a pure NE.*

Proposition 3.2 (Mixed NE on the All-k/Grim Face)**.** *For $M \geq 4$ and large T, if $p < k/3 = M/6$, then $\big((x_1^*,\, 0,\, 1-x_1^*),\ (x_1^*,\, 0,\, 1-x_1^*)\big)$ is a symmetric NE, where*

$$x_1^* = \frac{2(M-3+3p)}{3(M-2)}.$$

For $M = 4$: $x_1^* = (1+3p)/3$. This rest point is ***unstable****.*

Example 3.1 (NE for $M = 4$, $T = 10$)**.** *For $p = 3/4 > 2/3$: symmetric NE are $((0,0,1),(0,0,1))$ and $((0,1,0),(0,1,0))$; no mixed NE exists.*

For $p = 3/5 < 2/3$: symmetric NE are $((0,0,1),(0,0,1))$, $((1,0,0),(1,0,0))$, and $((x_1^*, 0, 1-x_1^*),(x_1^*, 0, 1-x_1^*))$ with $x_1^* = 14/15 \approx 0.933$.

3.3 Replicator Dynamics and Stability

The replicator equation is identical to Variation 1:

$$\dot{x}_m = x_m \left[(C\mathbf{x})_m - \mathbf{x}^\top C\mathbf{x}\right], \quad m = 1,2,3.$$

Proposition 3.3 (Rest Points)**.** *For $M = 4$, $k = 2$, and large T, the admissible rest points are:*

1. *The three pure vertices $(1,0,0)$, $(0,1,0)$, $(0,0,1)$.*
2. *The interior All-k/Grim rest point $(x_1^*, 0, 1-x_1^*)$ with $x_1^* = (1+3p)/3$, existing iff $p < 2/3$.*

*There are no interior rest points and no rest points on the All-*1*/Grim face.*

Proposition 3.4 (Stability of Pure Grim). *For $M \geq 4$ and all $p \in (\frac{1}{2}, 1]$, the rest point $(0, 0, 1)$ is asymptotically stable. The Jacobian eigenvalues are:*

$$\lambda_1 = C(1,3) - C(3,3) \approx T(3 - 3p - M) < 0, \quad \lambda_2 = C(2,3) - C(3,3) \approx T(1 - M) < 0.$$

Proposition 3.5 (Stability of Pure All-1). *For $M \geq 4$ and large T:*

1. *If $p > 2/3$, then $(0, 1, 0)$ is asymptotically stable.*
2. *If $p \leq 2/3$, then $(0, 1, 0)$ is unstable.*

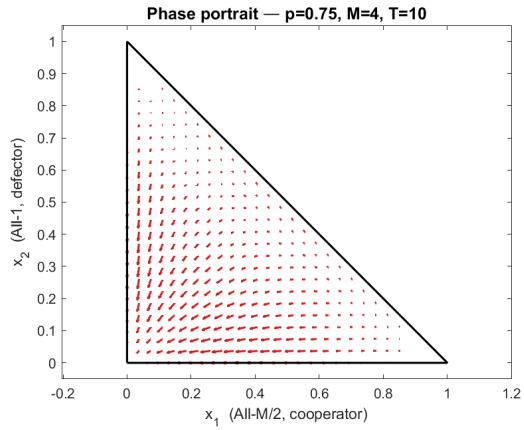
The eigenvalues are $\lambda_1 = T(2 - 3p)$ and $\lambda_3 = 2 - 3p$.

Proposition 3.6 (Stability of Pure All-k). *For $M = 4$, $k = 2$, and large T:*

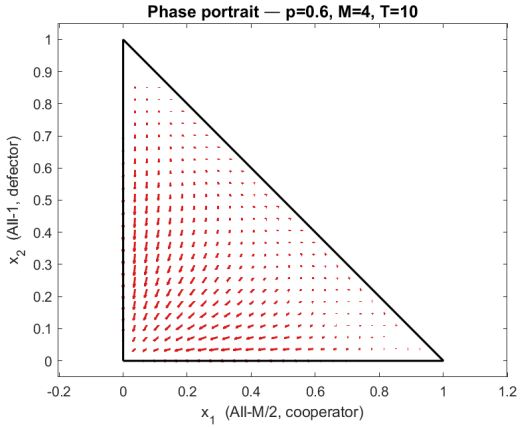
1. *If $p < 2/3$, then $(1, 0, 0)$ is asymptotically stable.*
2. *If $p \geq 2/3$, then $(1, 0, 0)$ is unstable.*

Proposition 3.7 (Mixed All-k/Grim Rest Point is Unstable). *For $M = 4$ and $p < 2/3$, the rest point $\mathbf{x}^* = (x_1^*, 0, 1 - x_1^*)$ is unstable. On the All-k/Grim face: $\dot{x}_1 = Tx_1(1 - x_1)(3x_1 - (1 + 3p))$, which is negative for $x_1 < x_1^*$ and positive for $x_1 > x_1^*$: a repeller on this edge.*

Phase Portraits and Sensitivity



(a) $p = 3/4$: $(0, 1, 0)$ and $(0, 0, 1)$ are stable.



(b) $p = 3/5$: $(1, 0, 0)$ and $(0, 0, 1)$ are stable; mixed rest point at $x_1^* \approx 0.933$.

Figure 17: Phase portraits, Variation 2 ($M = 4$, $T = 10$).

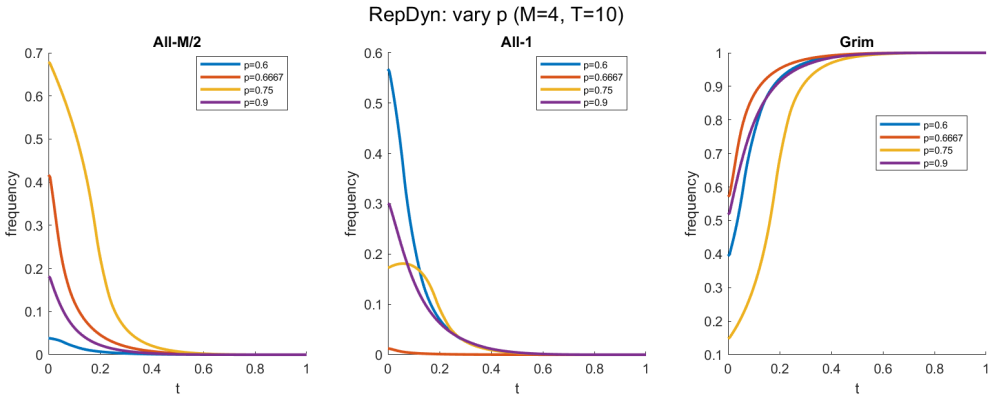


Figure 18: Replicator dynamics: effect of $p \in \{3/5, 2/3, 3/4, 9/10\}$ ($M = 4$, $T = 10$), Variation 2.

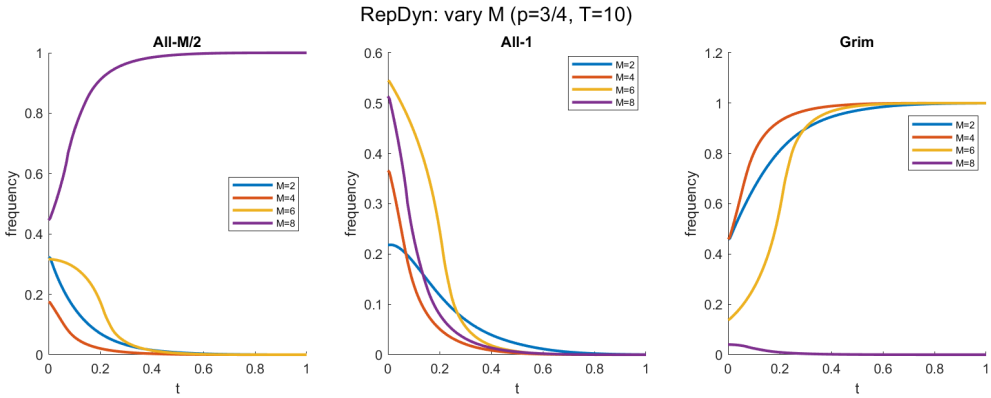


Figure 19: Replicator dynamics: effect of $M \in \{2, 4, 6, 8\}$ ($p = 3/4$, $T = 10$), Variation 2. For larger M, the threshold $p^* = M/6$ shifts.

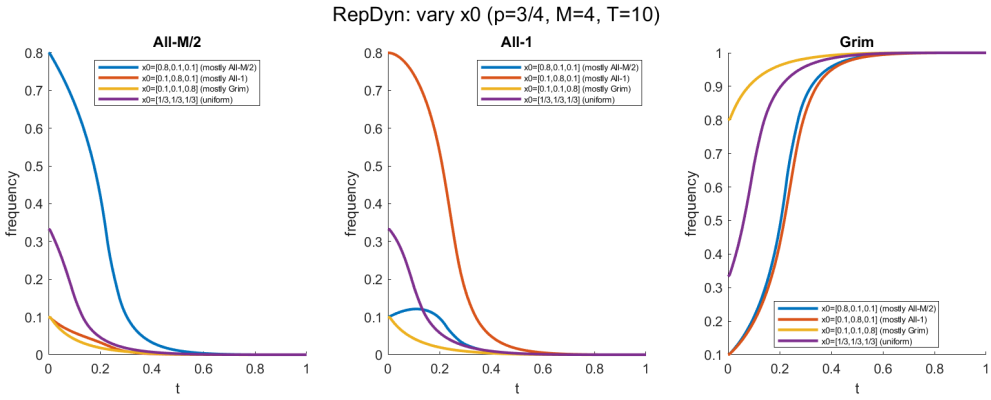


Figure 20: Replicator dynamics: initial condition sensitivity ($p = 3/4$, $M = 4$, $T = 10$), Variation 2.

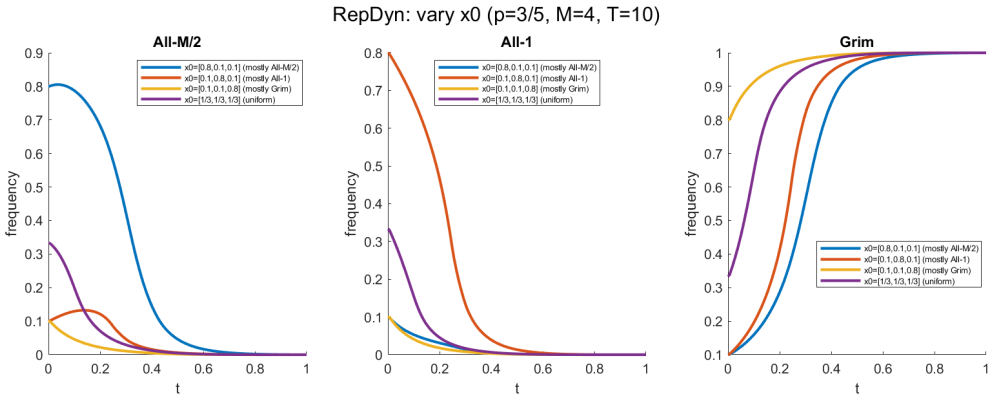


Figure 21: Replicator dynamics: initial condition sensitivity ($p = 3/5$, $M = 4$, $T = 10$), Variation 2.

3.4 Evolutionarily Stable Strategies

Definition 3.2 (ESS). *A mixed strategy* $\hat{\mathbf{x}}$ *is an* evolutionarily stable strategy *(ESS) iff for all* $\mathbf{x} \neq \hat{\mathbf{x}}$*:* $\hat{\mathbf{x}}^\top C\hat{\mathbf{x}} \geq \mathbf{x}^\top C\hat{\mathbf{x}}$*, and equality implies* $\hat{\mathbf{x}}^\top C\mathbf{x} > \mathbf{x}^\top C\mathbf{x}$*.*

Proposition 3.8 (Pure Grim is Always ESS). *For* $M \geq 4$ *and large* T*,* $(0, 0, 1)$ *is the unique ESS for all* p*. The deviation gain* $\hat{\mathbf{x}}^\top C\hat{\mathbf{x}} - \mathbf{x}^\top C\hat{\mathbf{x}} \approx aT(M-3+3p)+bT(M-1) > 0$ *for all* $\mathbf{x} \neq (0, 0, 1)$*.*

Proposition 3.9 (Conditional ESS).

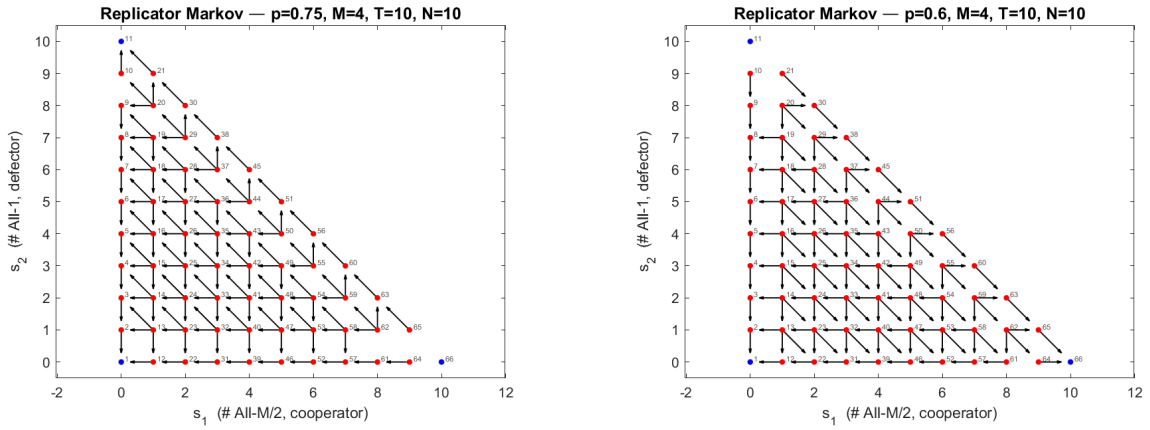
1. $(0, 1, 0)$ *is an ESS if and only if* $p > 2/3$*.*
2. $(1, 0, 0)$ *is an ESS if and only if* $p < 2/3$*.*
3. *The mixed rest point* $(x_1^*, 0, 1 - x_1^*)$ *is never an ESS (it is unstable).*

3.5 Markov Chain Analysis

The Markov chain model and PPI protocol are as in Variation 1.

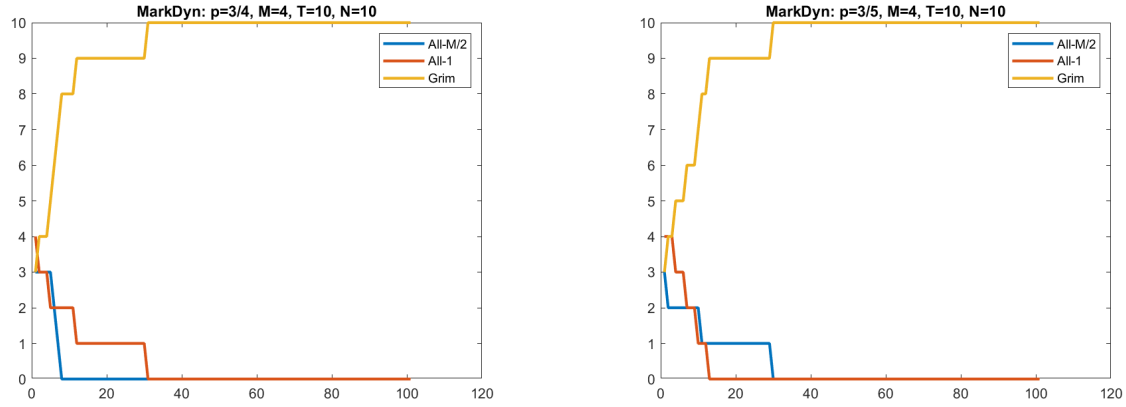
Proposition 3.10 (Absorbing States and Interior Transitions). *For* $M \geq 4$*,* $N \geq 4$*, and large* T*, the only absorbing states are* $(N, 0, 0)$*,* $(0, N, 0)$*,* $(0, 0, N)$*. In the interior, all transitions from All-*1 *go south/southeast (toward Grim or All-*k*), never north. The All-*k*–Grim direction is governed by:*

- $p > 2/3$*: west transitions (All-*k *→Grim); the* $s_2 = 0$ *face is absorbed entirely into* $(0, 0, N)$*.*
- $p < 2/3$*: east transitions (Grim→All-*k*); the* $s_2 = 0$ *face is separated by the line* $\ell : \; s_1(2 - 3p) = s_3(1 + 3p) - 2$*.*



(a) $p = 3/4$ (b) $p = 3/5$

Figure 22: Markov state transition graphs, Variation 2 ($M = 4$, $T = 10$, $N = 10$).



(a) $p = 3/4$ (b) $p = 3/5$

Figure 23: Markov dynamics time series, Variation 2 ($M = 4$, $T = 10$, $N = 10$, $s_0 = [3, 4, 3]$).

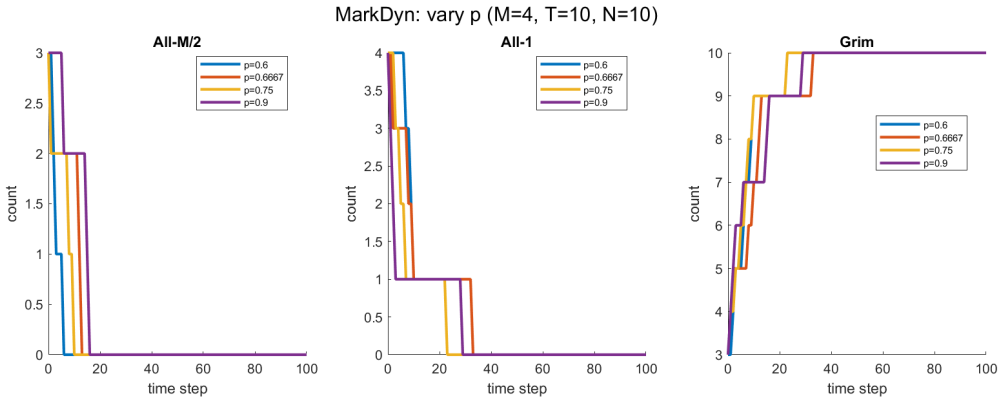


Figure 24: Markov dynamics: effect of $p \in \{3/5, 2/3, 3/4, 9/10\}$ ($M = 4$, $T = 10$, $N = 10$), Variation 2.

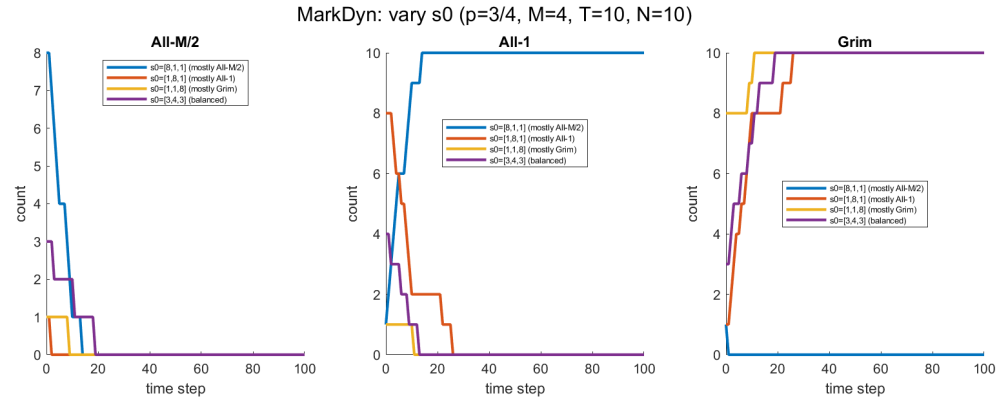


Figure 25: Markov dynamics: initial state sensitivity ($p = 3/4$, $M = 4$, $T = 10$, $N = 10$), Variation 2.

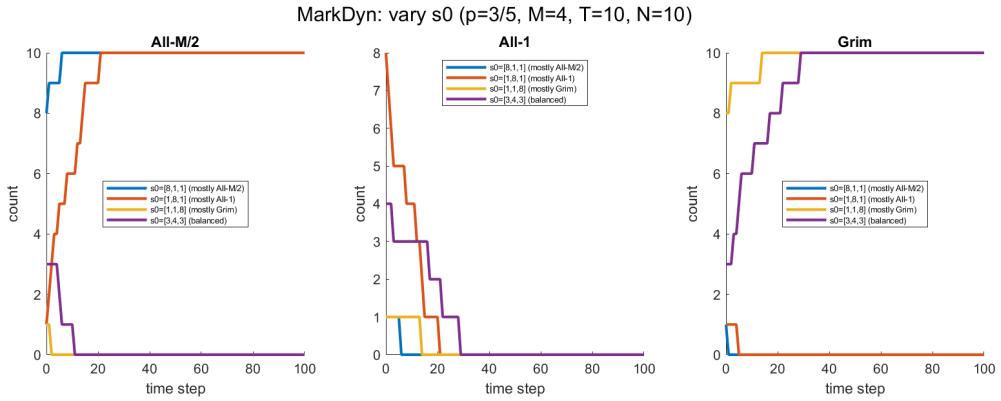


Figure 26: Markov dynamics: initial state sensitivity ($p = 3/5$, $M = 4$, $T = 10$, $N = 10$), Variation 2.

3.6 Connection to Learning Theory (Variations 1 & 2)

Both evolutionary game theory (EGT) and learning models describe adaptive behaviour in strategic environments and generate identical dynamics under distinct interpretations. In EGT, the population share $x_i(t)$ evolves via the replicator dynamic. In learning models, individual agents adjust mixed strategies based on reinforcement: the propensity q_i to play action i evolves as $q_i(t+1) = q_i(t) + \alpha\pi_i(t)$, with $x_i(t) = q_i(t)/\sum_j q_j(t)$. It can be shown [6] that in large populations and continuous time this rule converges to the replicator dynamic.

In the ISC context:

- **Evolutionary:** The fraction of players using Grim grows when Grim earns above-average payoffs; the stable outcomes represent population-level selection pressures governed by $p^* = 2/3$.
- **Learning:** Individual players accumulate experience and shift probability toward more profitable strategies; Grim's superior self-play payoff $C(3,3) = TM$ means rational learners reinforce cooperative reciprocal strategies over time.

4 Variation 3: Five-Strategy Extension — All-M, All-1, Grim, Grim-Patient, Grim-Forgiver

4.1 Overview and New Strategies

This variation restores the original All-M strategy and augments the strategy set with two new Grim variants, for a total of five strategies:

Definition 4.1 (Five ISC Strategies)**.**

1. σ_1 *(**All-**M): play ρ_M in every round.*
2. σ_2 *(**All-**1): play ρ_1 in every round.*
3. σ_3 *(**Grim**): play ρ_M until the opponent plays ρ_m with $m < M$; then switch permanently to ρ_1.*
4. σ_4 *(**Grim-Patient**): play ρ_M until the opponent plays ρ_m with $m < M$ for* two consecutive rounds*; then switch permanently to ρ_1.*
5. σ_5 *(**Grim-Forgiver**): play ρ_M; when the opponent plays ρ_m with $m < M$, switch to ρ_1 for* exactly two rounds*, then return unconditionally to ρ_M and repeat.*

Strategies σ_1, σ_3, σ_4, σ_5 all play ρ_M when the opponent plays ρ_M. Consequently, none of them ever triggers against each other; they all cooperate at ρ_M throughout any matchup among themselves.

4.2 The Five-Strategy Payoff Matrix

Let $n_3(T) = \lceil T/3 \rceil$.

Proposition 4.1 (ISC Payoff Matrix)**.**

$$C = \begin{pmatrix} TM & 3(1-p)T & TM & TM & TM \\ 3pT & T & 3p+(T-1) & 6p+(T-2) & 3p\,n_3+(T-n_3) \\ TM & 3(1-p)+(T-1) & TM & TM & TM \\ TM & 6(1-p)+(T-2) & TM & TM & TM \\ TM & 3(1-p)\,n_3+(T-n_3) & TM & TM & TM \end{pmatrix}. \tag{2}$$

Derivation sketch. All matchups among $\{1,3,4,5\}$ yield TM since no trigger is ever fired. Against All-1: Grim plays ρ_M in round 1 then triggers ($C(3,2) = 3(1-p)+(T-1)$); Grim-Patient plays ρ_M in rounds 1–2 then triggers ($C(4,2) = 6(1-p)+(T-2)$); Grim-Forgiver cycles with period 3 ($C(5,2) = 3(1-p)n_3 + (T-n_3)$). All-1 earns $3p$ per round against Grim's ρ_M rounds, etc. □

Remark 4.1 (Payoff degeneracy)**.** *The matrix C is highly degenerate: $C(i,j) = TM$ for all $i,j \in \{1,3,4,5\}$. This means all four cooperating strategies earn identical payoffs against each other. Moreover, $C(2,1) < C(2,3) < C(2,4) < C(2,5)$ always: Grim-Forgiver is the most exploitable by All-*1 *because it repeatedly returns to full cooperation. Conversely, $C(1,2) > C(3,2) > C(4,2) > C(5,2)$ for $p > 2/3$.*

Table 5: Payoff matrix C for $p = 3/4$, $M = 4$, $T = 10$, Variation 3.

	All-M	All-1	Grim	GrimP	GrimF
All-M	40	7.5	40	40	40
All-1	22.5	10	11.25	12.5	15
Grim	40	9.75	40	40	40
GrimP	40	9.5	40	40	40
GrimF	40	9	40	40	40

Table 6: Payoff matrix C for $p = 3/5$, $M = 4$, $T = 10$, Variation 3.

	All-M	All-1	Grim	GrimP	GrimF
All-M	40	12	40	40	40
All-1	18	10	10.8	11.6	13.4
Grim	40	10.2	40	40	40
GrimP	40	10.4	40	40	40
GrimF	40	11	40	40	40

4.3 Nash Equilibria

Proposition 4.2 (Pure-Strategy NE)**.** *For $M \geq 4$ and large T:*

1. *All pure states (σ_i, σ_i) for $i \in \{1, 3, 4, 5\}$ are NE.*
2. *(σ_2, σ_2) is a NE if and only if $p \geq \frac{2}{3}$.*

Proposition 4.3 (Cooperative Equilibrium Set)**.** *For every $M \geq 4$ and large T, all mixed strategies of the form $\mathbf{x} = (x_1, 0, x_3, x_4, x_5)$ with $x_1 + x_3 + x_4 + x_5 = 1$ are Nash equilibria. The ISC has a* continuum *of NE due to the payoff degeneracy $C(i, j) = TM$ for all $i, j \in \{1, 3, 4, 5\}$.*

4.4 Replicator Dynamics

Let $\mathbf{x} = (x_1, x_2, x_3, x_4, x_5)$ with $\sum_m x_m = 1$. The replicator equation is:

$$\dot{x}_m = x_m \Big[(C\mathbf{x})_m - \mathbf{x}^\top C\mathbf{x} \Big], \quad m = 1, \ldots, 5.$$

Proposition 4.4 (Rest Points)**.** *The admissible rest points are:*

1. *The entire cooperative face $\mathcal{F} = \{\mathbf{x} \in S_5 : x_2 = 0\}$.*
2. *The pure All-1 state $(0, 1, 0, 0, 0)$.*
3. *For $p \geq \frac{2}{3}$, all mixtures between All-1 and any cooperating profile.*

Within $\mathcal{F}$, the replicator dynamics has no drift *— there is no selection pressure among cooperating strategies.*

Proposition 4.5 (Stability of $\mathcal{F}$)**.** *For $M \geq 4$ and large T:*

1. *If $p < \frac{2}{3}$: $\mathcal{F}$ is asymptotically stable (All-1 cannot invade).*
2. *If $p > \frac{2}{3}$: $\mathcal{F}$ is unstable (All-1 invades).*
3. *If $p = \frac{2}{3}$: $\mathcal{F}$ is unstable (Grim variants invade).*

The eigenvalue for All-1 invasion is $\mu_m = C(m, 2) - T$, which is positive (cooperative strategy invades All-1) iff $p < 2/3$ for all $m \in \{1, 3, 4, 5\}$.

Proposition 4.6 (Grim Variant Ranking Under All-1 Invasion)**.** *For $p > 2/3$, when the population is near $\mathcal{F}$ with a small fraction ε of All-1 invaders, the fitness ranking is $f_3 > f_4 > f_5 > f_1$: Grim is most resistant to exploitation, and Grim-Forgiver is least resistant (though more resistant than All-M).*

Pentagon Phase Portrait

Because S_5 is four-dimensional, we represent dynamics using a *pentagon projection*: each strategy σ_m is placed at vertex m of a regular pentagon, and a population state is projected to $(X, Y) = \sum_m x_m(\cos\theta_m, \sin\theta_m)$ where $\theta_m = \pi/2 - 2\pi(m-1)/5$.

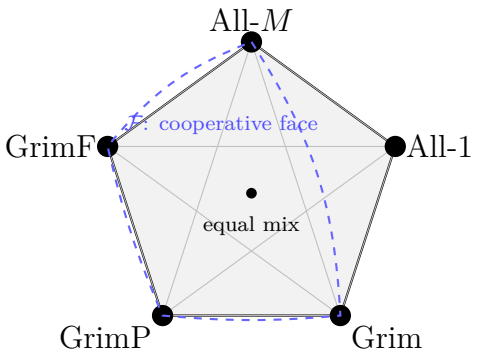


Figure 27: Pentagon representation of the five-strategy simplex S_5. The dashed region $\mathcal{F}$ is the cooperative face; all points in $\mathcal{F}$ are rest points of the replicator dynamics.

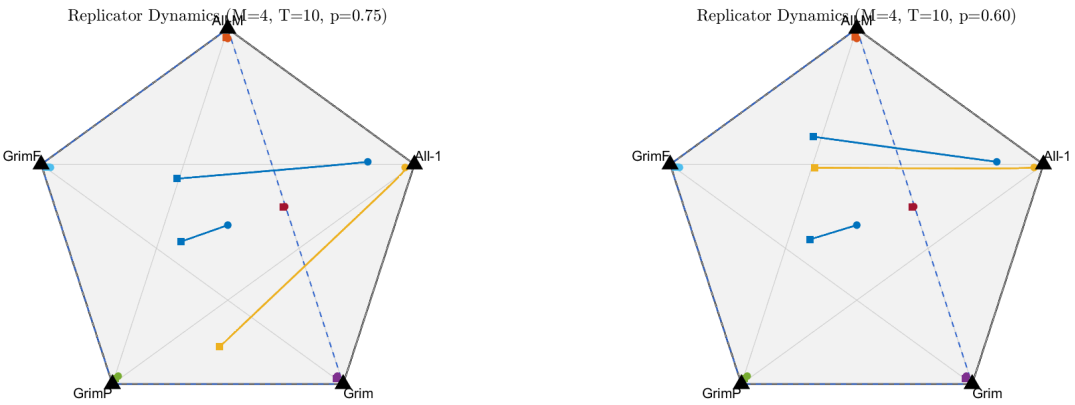
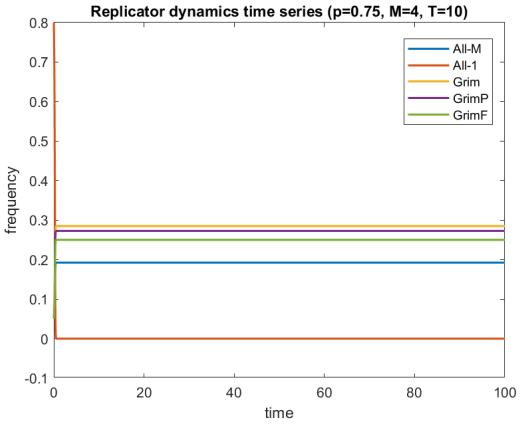
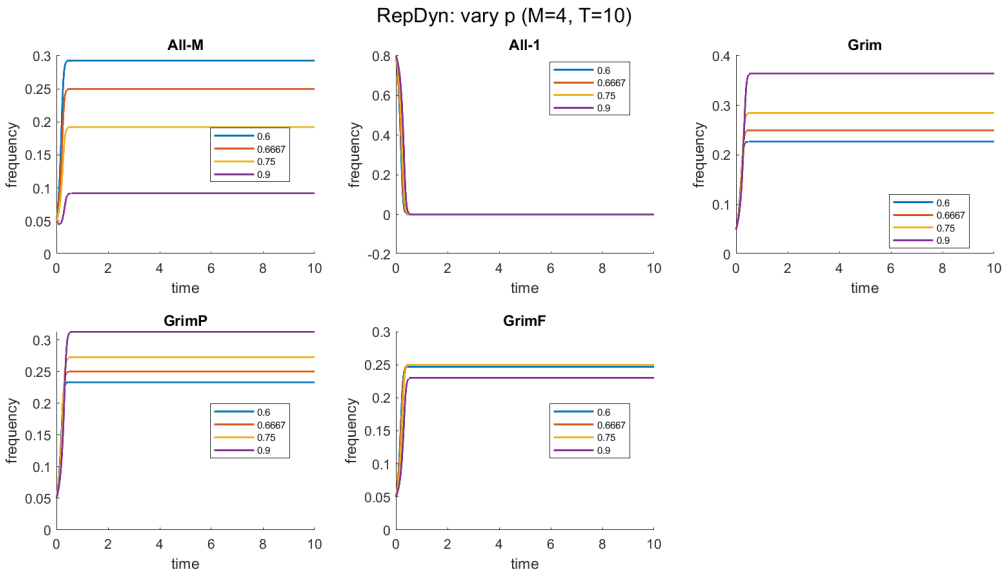


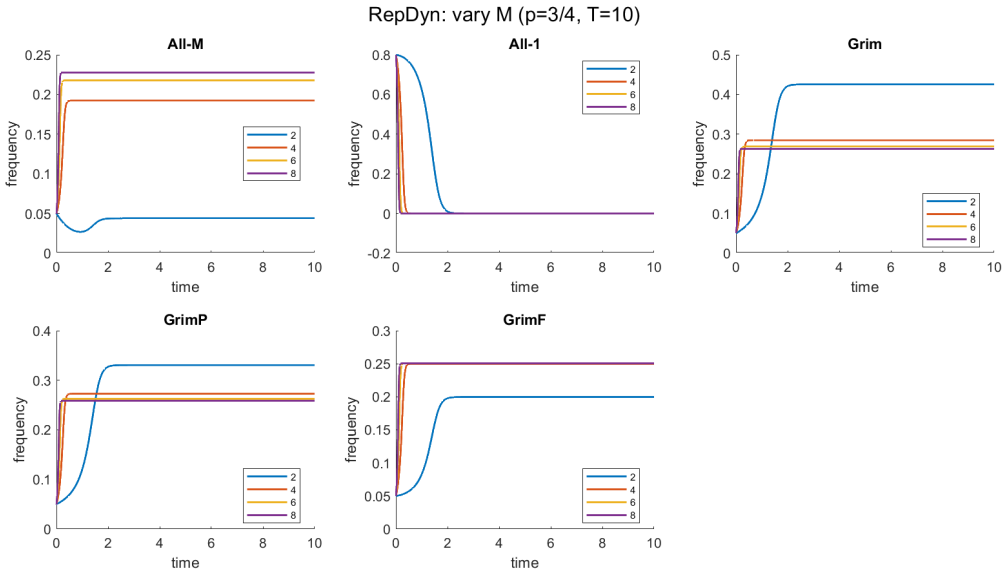
Figure 28: Phase portraits on the pentagon, Variation 3 ($M = 4$, $T = 10$).

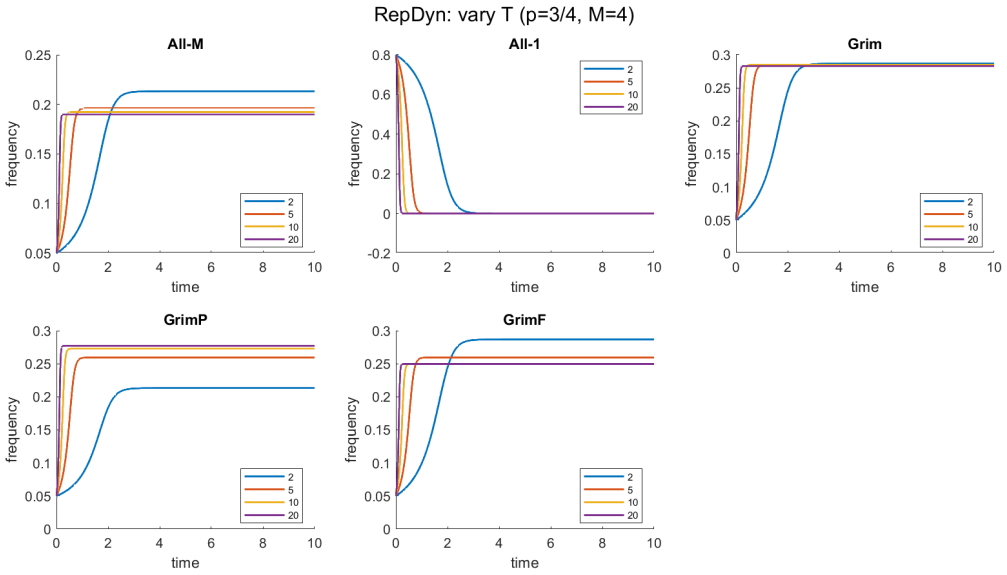


(a) $p = 3/5$.

Figure 29: Replicator dynamics time series, Variation 3 ($M = 4$, $T = 10$). Line colours: All-M blue, All-1 red, Grim green, GrimP orange, GrimF purple.







4.5 Evolutionary Stable Strategies

Proposition 4.7 (No Pure ESS from Cooperating Strategies)**.** *For $M \geq 4$ and large T, no pure strategy σ_i with $i \in \{1, 3, 4, 5\}$ is an ESS, because any other cooperating strategy σ_j ($j \neq i$, $j \in \{1, 3, 4, 5\}$) earns exactly the same payoff TM as σ_i against σ_i, violating the strict ESS condition.*

Proposition 4.8 (All-1 ESS Condition)**.** *For $M \geq 4$ and large T:*

1. *If $p > \frac{2}{3}$: $(0, 1, 0, 0, 0)$ is an ESS.*
2. *If $p \leq \frac{2}{3}$: $(0, 1, 0, 0, 0)$ is not an ESS.*

Proposition 4.9 (No Interior ESS)**.** *No mixed strategy in the interior of $\mathcal{F}$ is an ESS, because points in $\mathcal{F}$ are neutral to perturbations within $\mathcal{F}$.*

4.6 Markov Chain Analysis

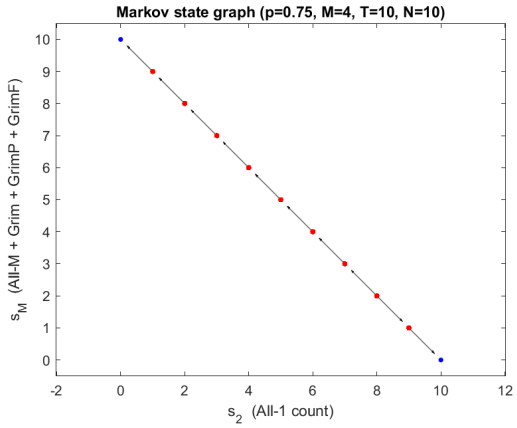
The population state is $\mathbf{s} = (s_1, s_2, s_3, s_4, s_5)$ with $\sum_m s_m = N$.

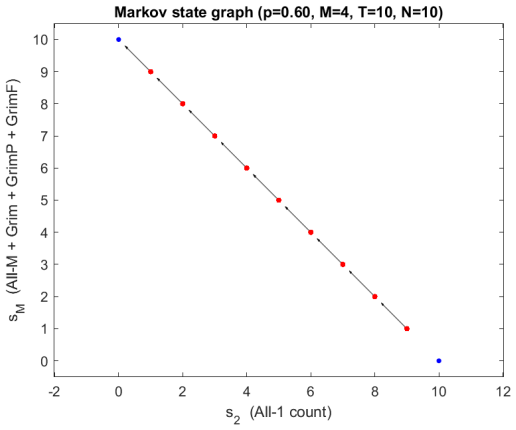
Proposition 4.10 (Absorbing States)**.** *For $M \geq 4$, $N \geq 2$, and large T, the absorbing states are:*

1. *The pure All-1 state $(0, N, 0, 0, 0)$.*
2. *The entire cooperative face: $\mathcal{A} = \{(s_1, 0, s_3, s_4, s_5) : s_1 + s_3 + s_4 + s_5 = N\}$.*

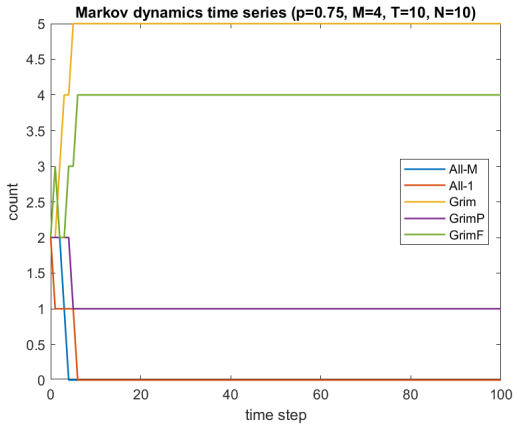
There are $\binom{N+3}{3} + 1$ absorbing states in total.

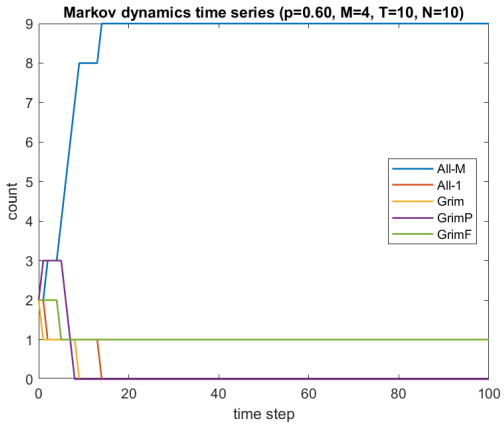
This is a significant generalisation of the original paper: the entire $s_2 = 0$ four-dimensional face is absorbing, because all four cooperating strategies earn identical payoffs TM against each other.



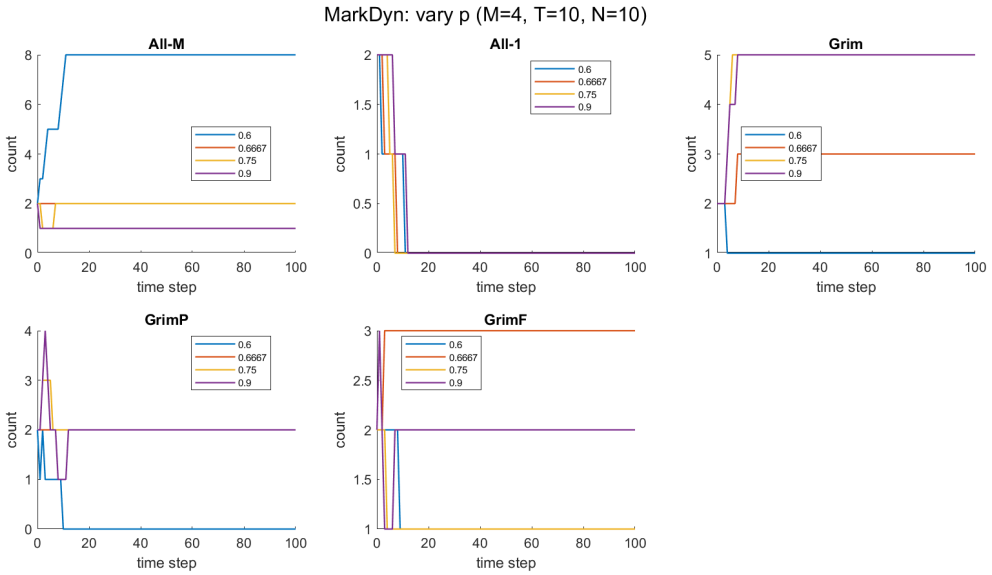


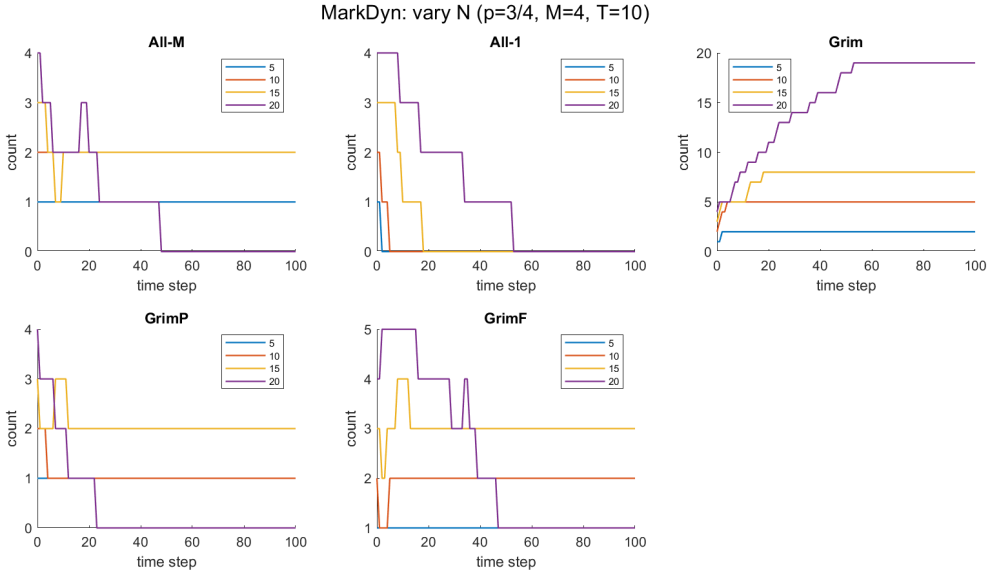
(a) $p = 3/4$.

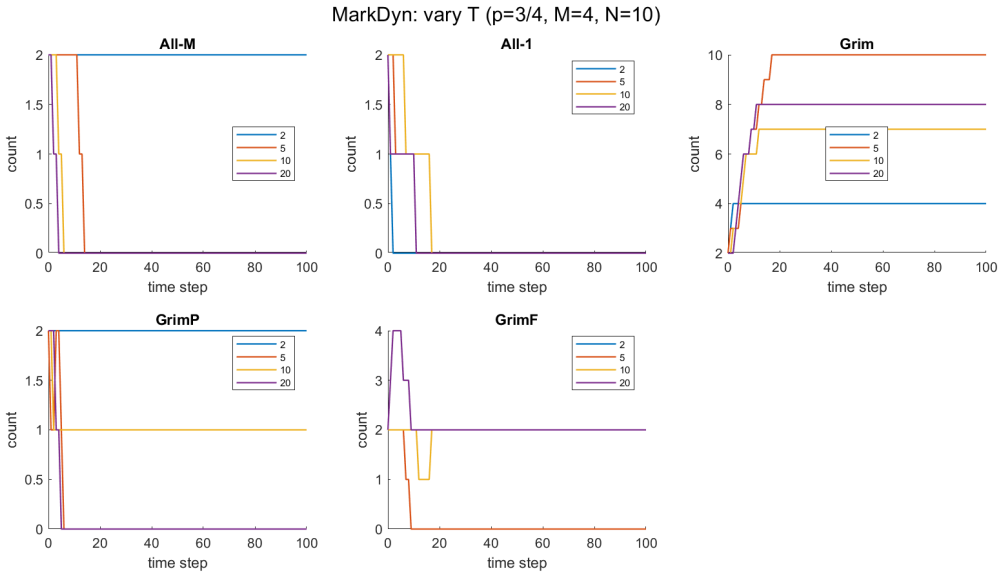


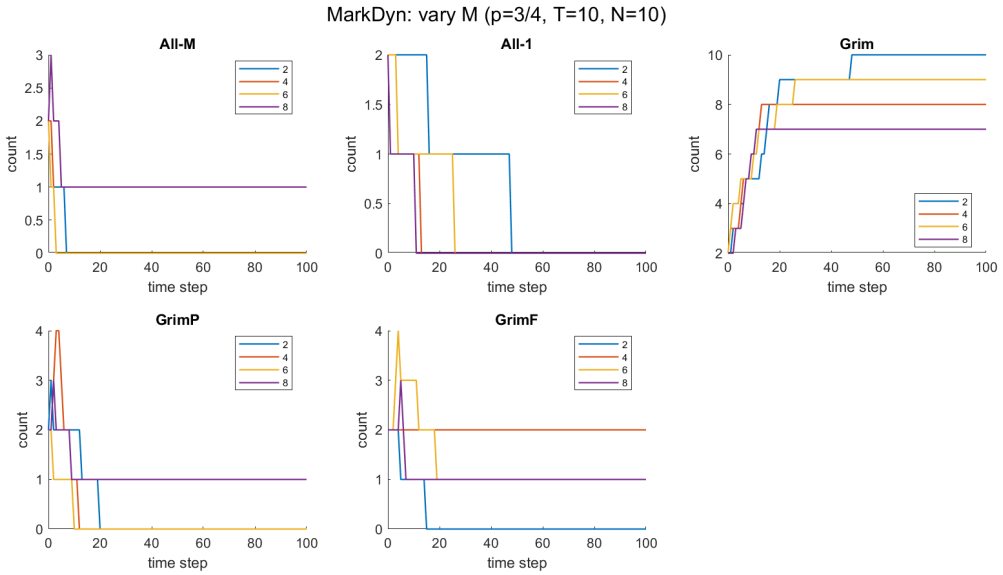


(a) $p = 3/4$.


MarkDyn: vary p (M=4, T=10, N=10)
All-M
All-1
Grim
GrimP
GrimF
count
time step
0.6
0.6667
0.75
0.9


MarkDyn: vary N (p=3/4, M=4, T=10)
All-M
All-1
Grim
GrimP
GrimF
count
time step
5
10
15
20





4.7 Connection to Learning Theory (Variation 3)

In the five-strategy ISC context:

- **Evolutionary.** The four cooperating strategies are *evolutionarily neutral* relative to each other on $\mathcal{F}$. Natural selection acts only on the All-1/cooperator axis, with threshold $p^* = 2/3$.
- **Learning.** A rational learner will reinforce any cooperating strategy over All-1 when $p < 2/3$ (cooperation yields TM vs. T). Among cooperating strategies, reinforcement is indifferent; the internal mix is determined by initial conditions.
- **Finite-population.** Under PPI, the absorbing face $\mathcal{A}$ captures the long-run state where social learning has eliminated All-1.

5 Variation 4: Two-Population Framework — All-M, All-1, and Grim

5.1 Overview and Motivation

This variation replaces the single evolving population with *two* populations that play exclusively against each other (no within-population games). Two key motivations drive this extension:

1. One-population replicator dynamics can only converge to *symmetric* Nash equilibria [5]. The ISC possesses asymmetric NE (e.g. Population 1 all All-M, Population 2 all Grim) that are inaccessible to one-population dynamics.

2. Many real interactions are inter-group: buyers vs. sellers, incumbents vs. entrants, immune systems vs. pathogens.

The strategy set is $\{\sigma_1 = \text{All-}M,\ \sigma_2 = \text{All-1},\ \sigma_3 = \text{Grim}\}$ for both populations. The bimatrix game is $(C, C^\top)$, where C is the same 3×3 matrix as in [1]:

$$C = \begin{pmatrix} TM & 3(1-p)\,T & TM \\ 3p\,T & T & 3p + (T-1) \\ TM & 3(1-p) + (T-1) & TM \end{pmatrix}. \tag{3}$$

Table 7: Payoff matrix C for $p = 3/4$, $M = 4$, $T = 10$, Variation 4. Population 2 payoff matrix is $C^\top$.

	vs All-M	vs All-1	vs Grim
All-M	40	7.5	40
All-1	22.5	10	11.25
Grim	40	9.75	40

Table 8: Payoff matrix C for $p = 3/5$, $M = 4$, $T = 10$, Variation 4.

	vs All-M	vs All-1	vs Grim
All-M	40	12	40
All-1	18	10	10.8
Grim	40	10.2	40

5.2 Nash Equilibria

Definition 5.1 (Two-Population Nash Equilibrium)**.** *A pair of mixed strategies $(\hat{x}, \hat{y})$ with $\hat{x}, \hat{y} \in \Delta_3$ is a NE of $(C, C^\top)$ iff $\hat{x}^\top C \hat{y} \geq x^\top C \hat{y}$ for all x and $\hat{y}^\top C^\top \hat{x} \geq y^\top C^\top \hat{x}$ for all y.*

Proposition 5.1 (Asymmetric NE Family). *For every $M \in \{3, 4, \ldots\}$, large T, and any $a', a'' \in [0, 1]$:*

$$\big((1 - a',\, 0,\, a'),\ (1 - a'',\, 0,\, a'')\big)$$

is a NE of the two-population ISC. This includes genuinely asymmetric *NE such as (e_1, e_3) (Pop 1 all All-M, Pop 2 all Grim).*

Proposition 5.2 (Symmetric NE). *The symmetric NE $(\hat{x}, \hat{x})$ of the two-population ISC coincide exactly with the NE of the one-population game: all $(1-a', 0, a')$ profiles, $((0, 1, 0), (0, 1, 0))$ iff $p \geq 2/3$, and the mixed NE from [1].*

Example 5.1 ($M = 4$, $T = 10$, $p = 3/4$). *NE include: all pairs $((1 - a', 0, a'), (1 - a'', 0, a''))$ for $a', a'' \in [0, 1]$, plus the symmetric mixed NE $\big((\frac{1}{8}, \frac{7}{8}, 0), (\frac{1}{8}, \frac{7}{8}, 0)\big)$.*

5.3 Two-Population Replicator Dynamics

Let $x(t) = (x_1, x_2, x_3)$ and $y(t) = (y_1, y_2, y_3)$ be the strategy frequency vectors of Population 1 and Population 2. Fitness of strategy m in Population 1 is $(Cy)_m$; in Population 2 it is $(C^\top x)_m$. The two-population replicator equations are:

$$\dot{x}_m = x_m\big((Cy)_m - x^\top Cy\big), \tag{4}$$

$$\dot{y}_m = y_m\big((x^\top C)_m - x^\top Cy\big). \tag{5}$$

The system conserves the simplices Δ_3^x and Δ_3^y separately. Setting $y = x$ recovers the standard one-population replicator.

Proposition 5.3 (Cooperative Face Always Stable). *For every $M \geq 4$ and large T, the entire face $\mathcal{F} = \{(x, y) : x_2 = y_2 = 0\}$ is stable in both transverse directions. The eigenvalue for All-1 invasion of Population 1 is:*

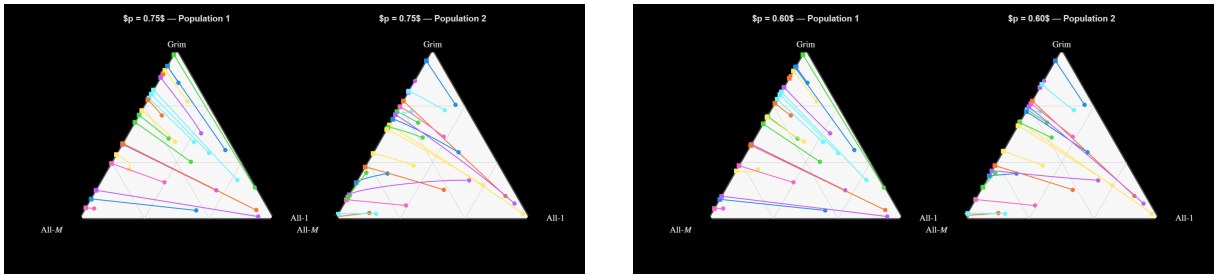
$$\lambda_{x_2} \approx T[1 - M + (3p - 1)b] \leq -1 < 0,$$

for $y^* = (1 - b, 0, b)$ and $M \geq 4$. Similarly for Population 2. This is stronger *than the one-population result (where $\mathcal{F}$ was only neutrally stable).*

Proposition 5.4 (Threshold Preserved). *The critical threshold $p^* = 2/3$ for All-1 stability is preserved:*

- *$p > 2/3$: All-1 is stable in both populations simultaneously.*
- *$p < 2/3$: All-1 is unstable; the system converges to $\mathcal{F}$.*

Phase Portraits and Sensitivity



(a) $p = 3/4$. (b) $p = 3/5$.

Figure 32: Phase portraits of the two-population replicator dynamics, Variation 4 ($M = 4$, $T = 10$). Solid: Population 1. Dashed: Population 2.

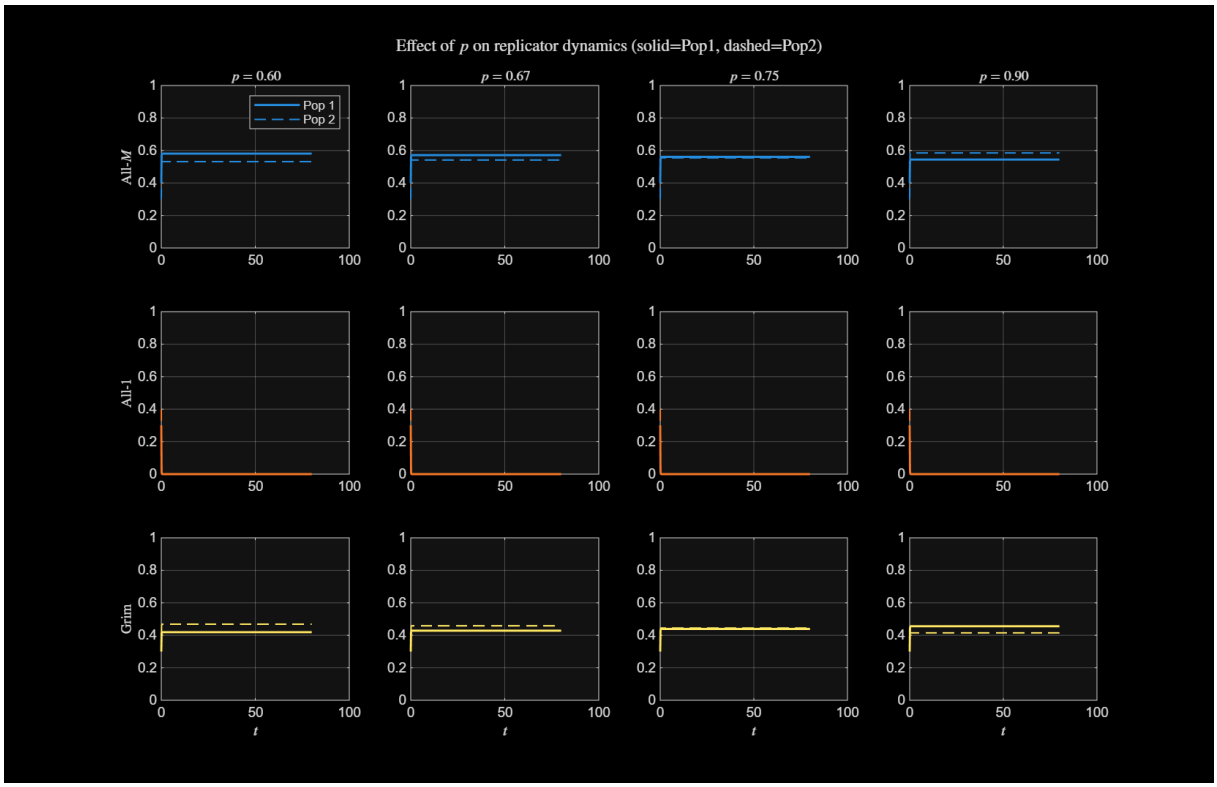


Figure 33: Effect of $p \in \{3/5, 2/3, 3/4, 9/10\}$ on two-population replicator dynamics, Variation 4 ($M = 4$, $T = 10$).

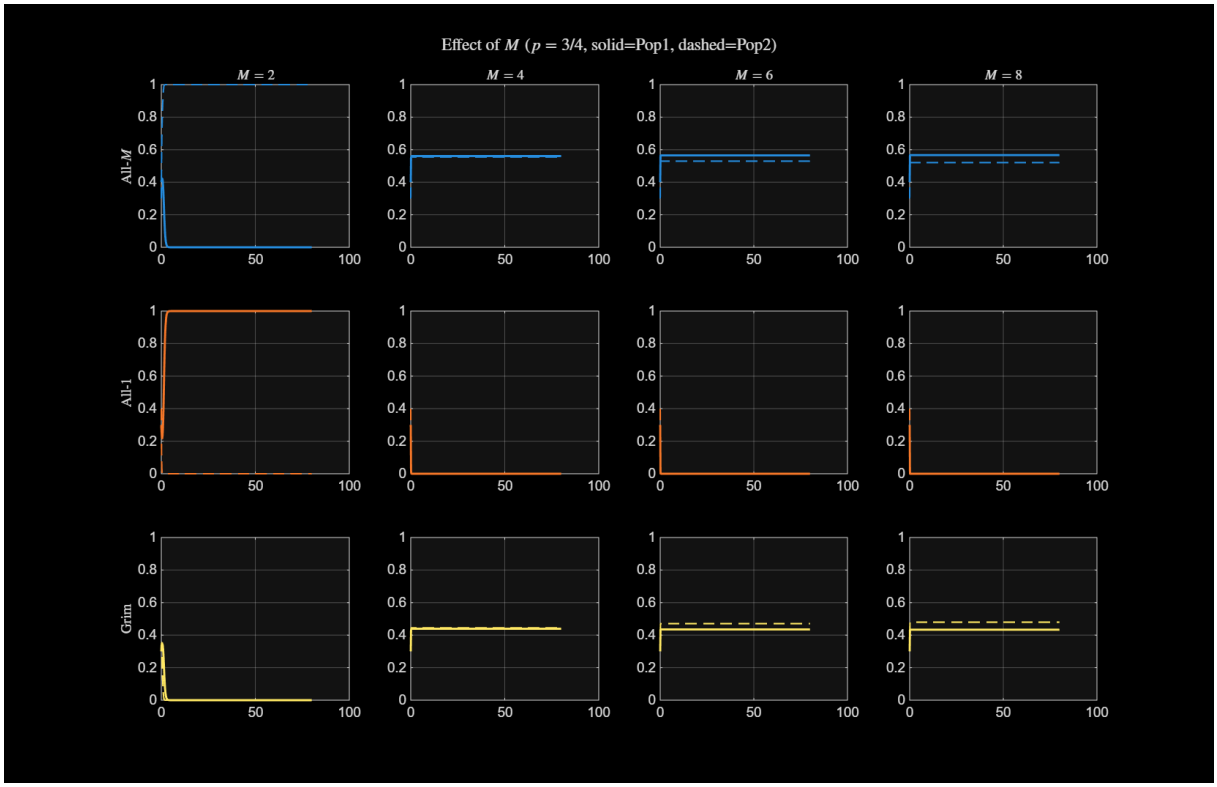


Figure 34: Effect of $M \in \{2, 4, 6, 8\}$ on two-population replicator dynamics, Variation 4 ($p = 3/4$, $T = 10$).

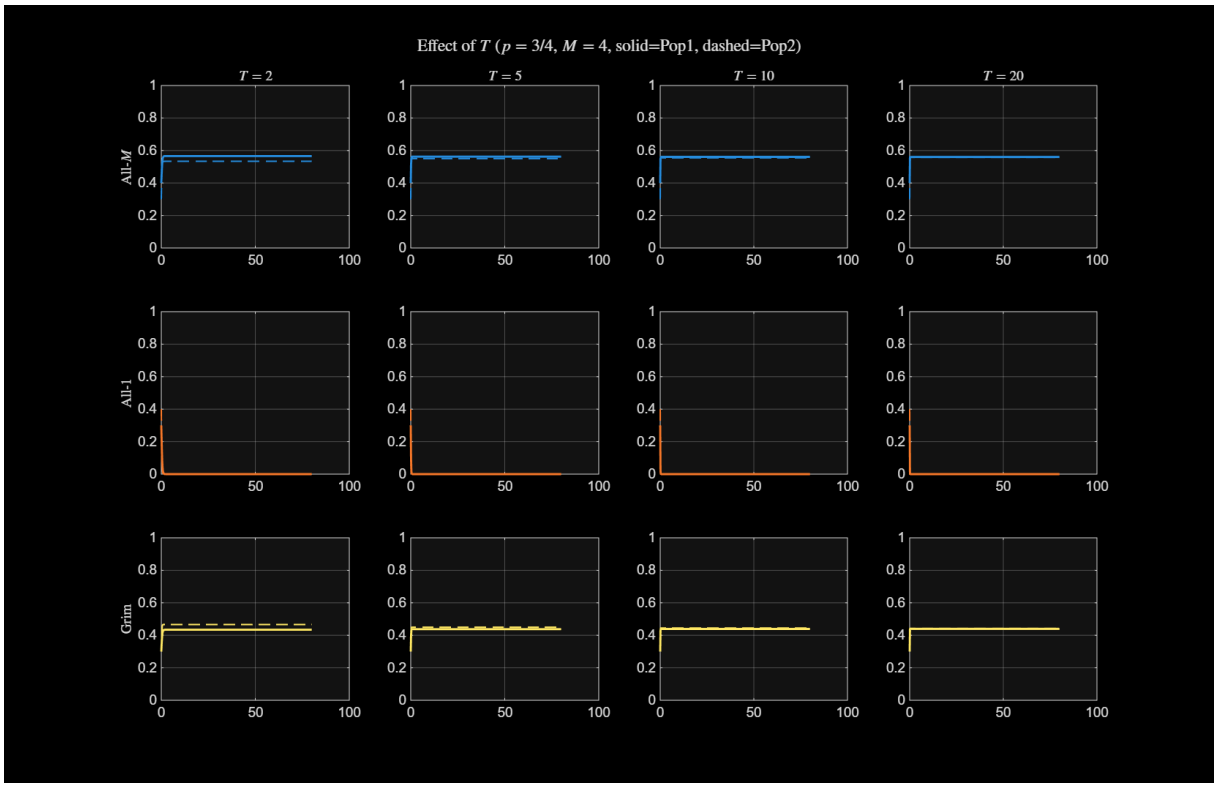


Figure 35: Effect of $T \in \{2, 5, 10, 20\}$ on two-population replicator dynamics, Variation 4 ($p = 3/4$, $M = 4$).

5.4 Evolutionary Stable Strategies

Proposition 5.5 (Two-Population ESS). *1. All-1 is the unique two-population ESS, iff $p > 2/3$.*

2. Asymmetric NE such as (e_1, e_3) are not *two-population ESS, because the strict ESS condition fails at equal payoffs $TM = TM$.*

3. No interior point of $\mathcal{F}$ is an ESS.

5.5 Two-Population Markov Chain

The population state is $(\mathbf{s}, \mathbf{t})$ where $\mathbf{s} \in \mathbb{Z}^3_{\geq 0}$ with $\sum_m s_m = N_1$ and $\mathbf{t}$ with $\sum_n t_n = N_2$. Per-player payoffs:

$$q_m^{(1)}(\mathbf{s}, \mathbf{t}) = \sum_n t_n \, C(m, n), \tag{6}$$

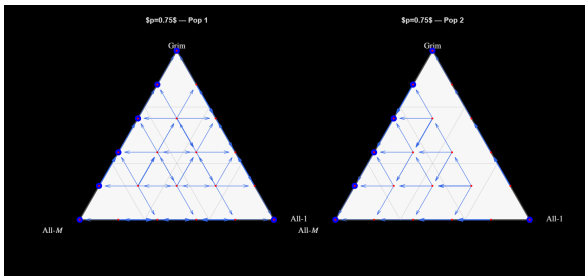
$$q_n^{(2)}(\mathbf{s}, \mathbf{t}) = \sum_m s_m \, C(m, n). \tag{7}$$

Proposition 5.6 (Absorbing States). *For $M \geq 4$, $N_1, N_2 \geq 2$, and large T, the absorbing states are:*

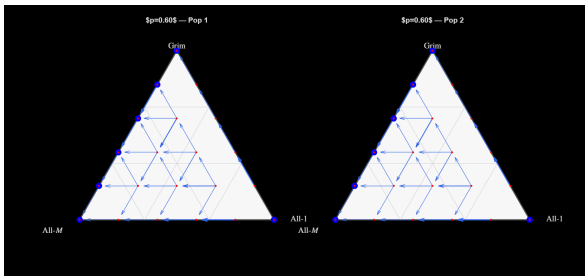
1. All states $(\mathbf{s}, \mathbf{t})$ with $s_2 = t_2 = 0$: $\mathcal{A} = \{(\mathbf{s}, \mathbf{t}) : s_1 + s_3 = N_1,\ t_1 + t_3 = N_2\}$. This face has $(N_1 + 1)(N_2 + 1)$ states.

2. The pure All-1 state $((0, N_1, 0), (0, N_2, 0))$.

The absorbing face $\mathcal{A}$ is *two-dimensional* (independent cooperation levels of the two populations), in contrast to the one-dimensional $s_2 = 0$ face of the one-population model.

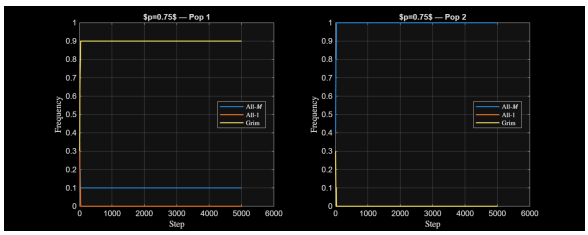


(a) $p = 3/4$.

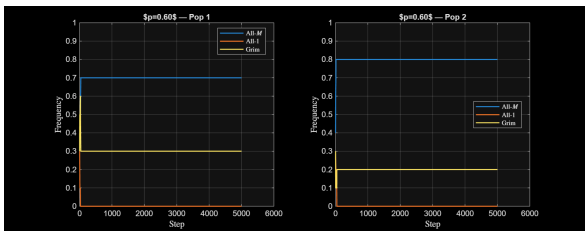


(b) $p = 3/5$.

Figure 36: Two-population Markov state transition graphs, Variation 4 ($M = 4$, $T = 10$, $N_1 = N_2 = 5$). Blue: absorbing states. Red: transient.



(a) $p = 3/4$.



(b) $p = 3/5$.

Figure 37: Two-population Markov dynamics time series, Variation 4 ($M = 4$, $T = 10$, $N_1 = N_2 = 10$, 5000 steps, $\mathbf{s}_0 = (3, 4, 3)$, $\mathbf{t}_0 = (4, 3, 3)$). Solid: Pop 1. Dashed: Pop 2.

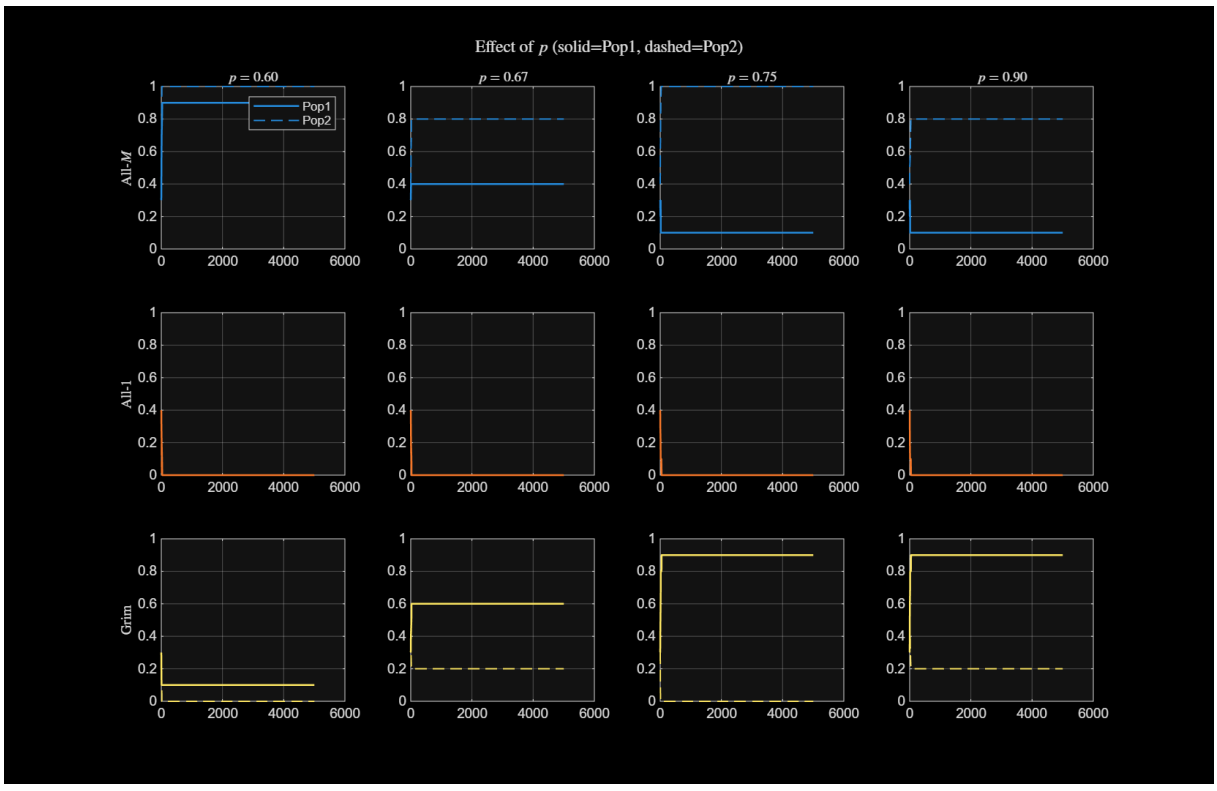


Figure 38: Effect of $p \in \{3/5, 2/3, 3/4, 9/10\}$, Variation 4 ($M = 4$, $T = 10$, $N = 10$).

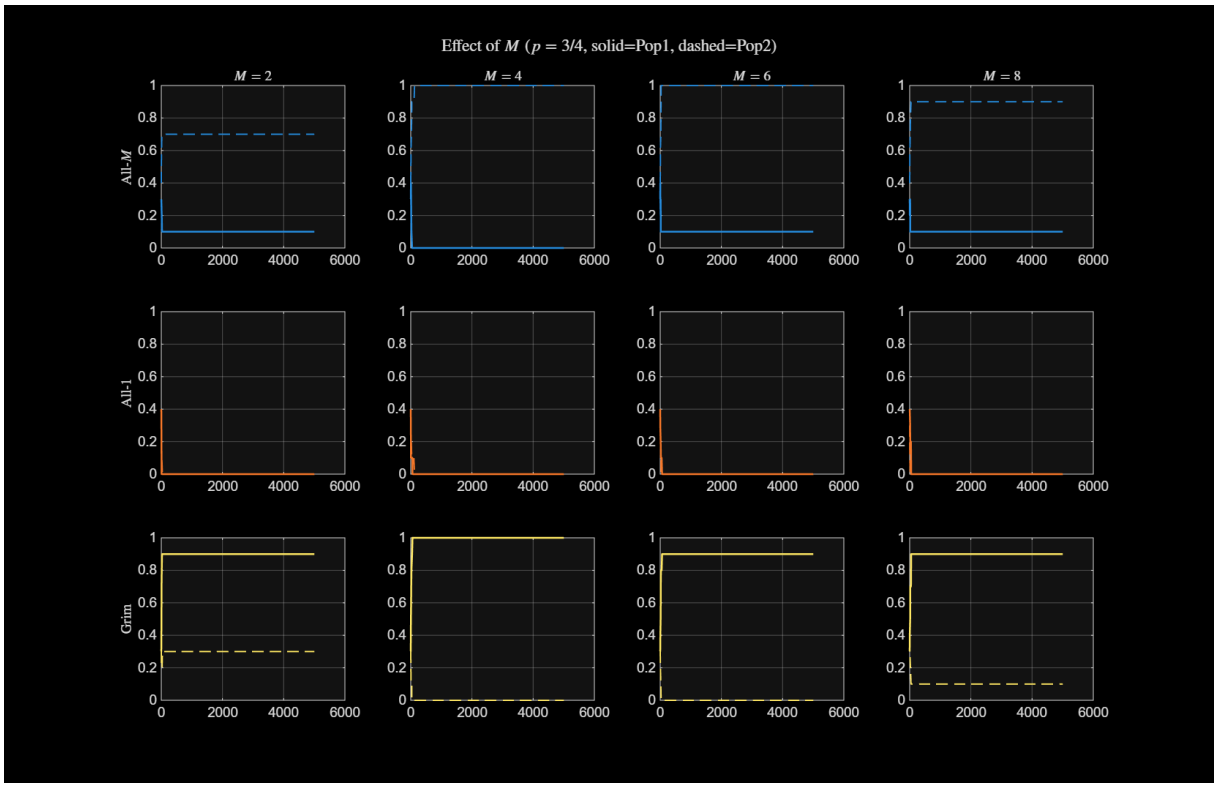


Figure 39: Effect of $M \in \{2, 4, 6, 8\}$, Variation 4 ($p = 3/4$, $T = 10$, $N = 10$).

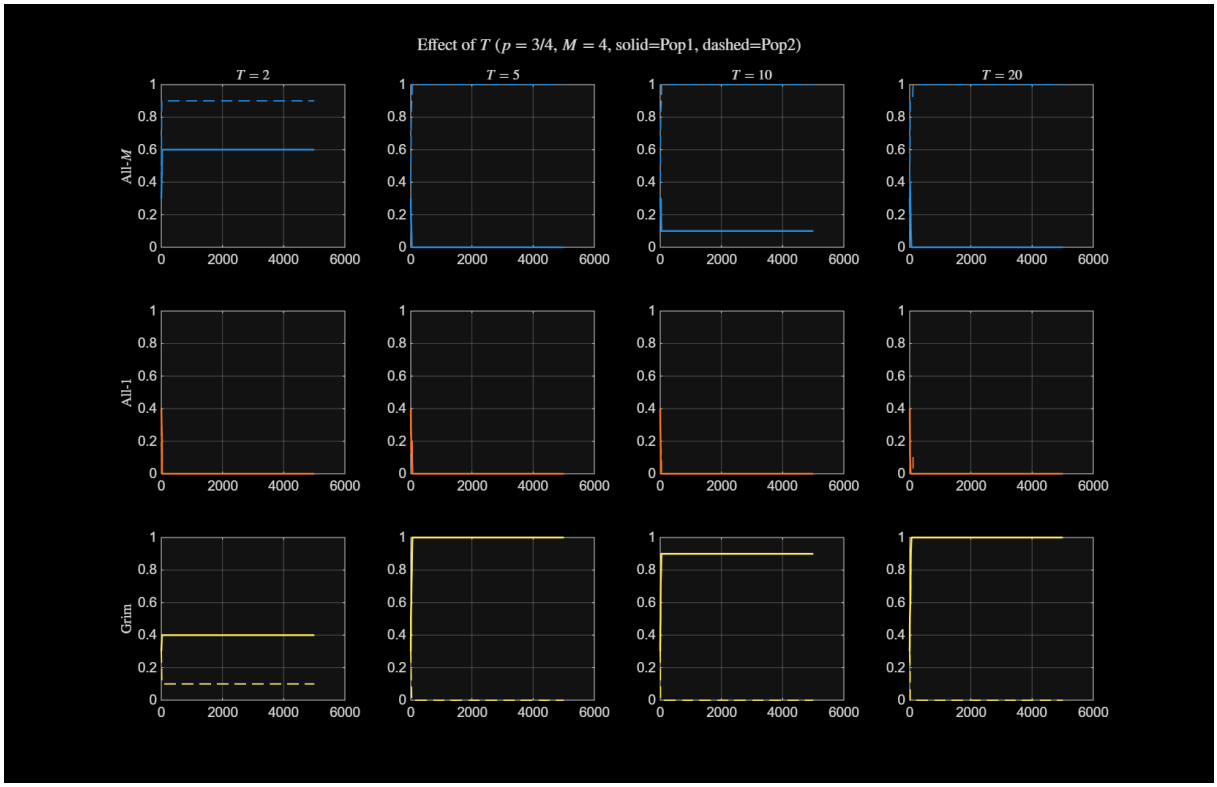


Figure 40: Effect of $T \in \{2, 5, 10, 20\}$, Variation 4 ($p = 3/4$, $M = 4$, $N = 10$).

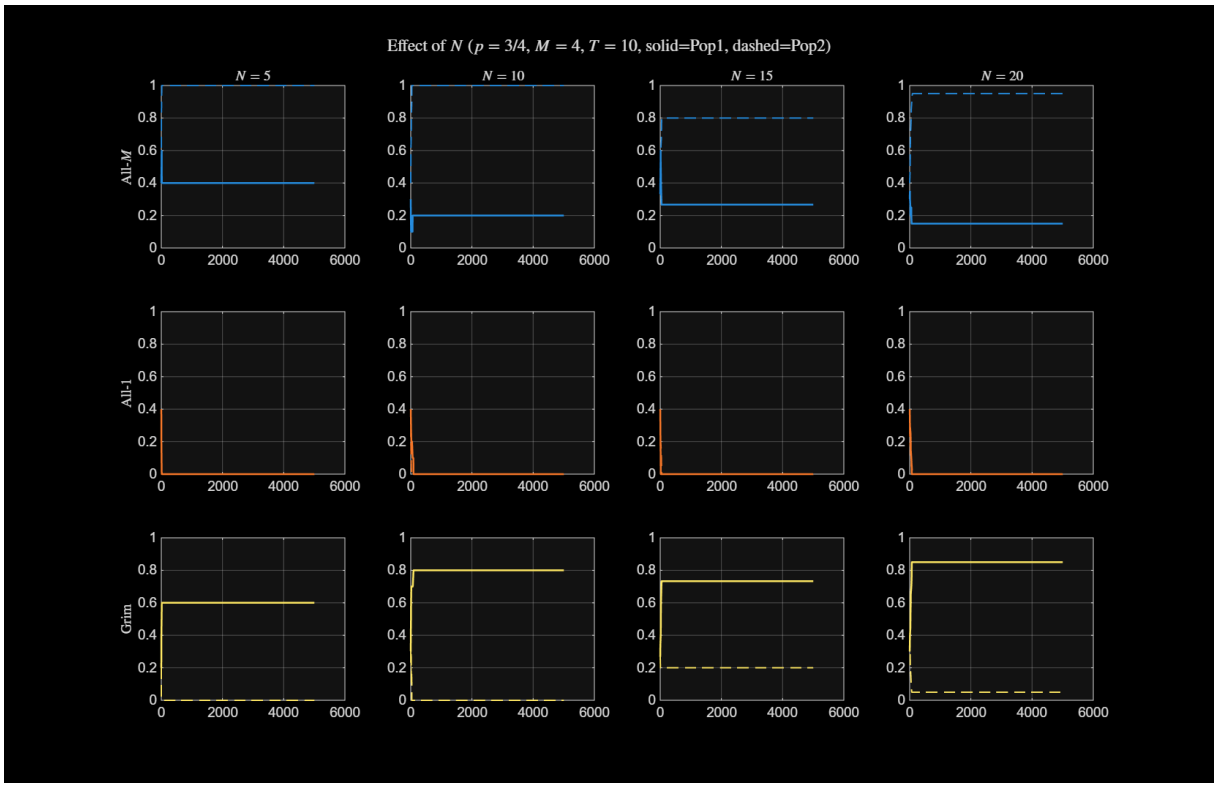


Figure 41: Effect of $N \in \{5, 10, 15, 20\}$ ($N_1 = N_2 = N$), Variation 4 ($p = 3/4$, $M = 4$, $T = 10$).

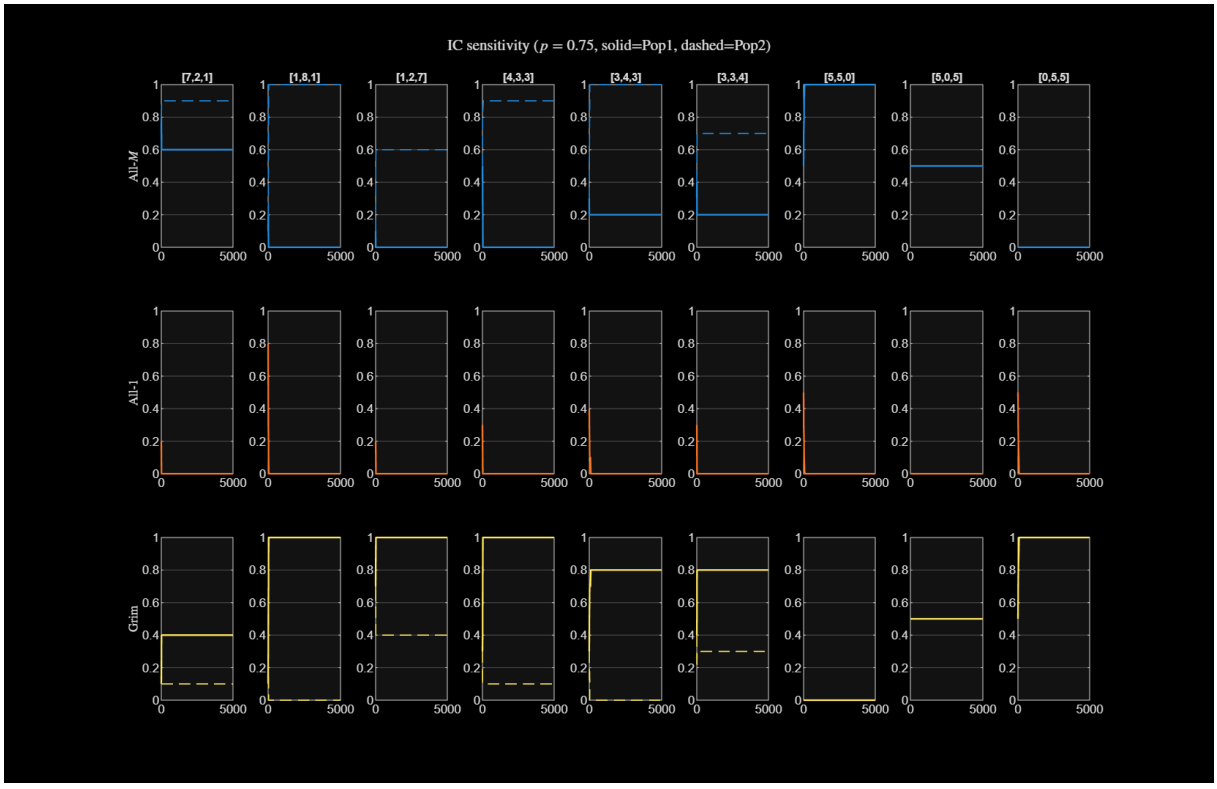


Figure 42: Initial state sensitivity ($p = 3/4$, $M = 4$, $T = 10$, $N = 10$), Variation 4.

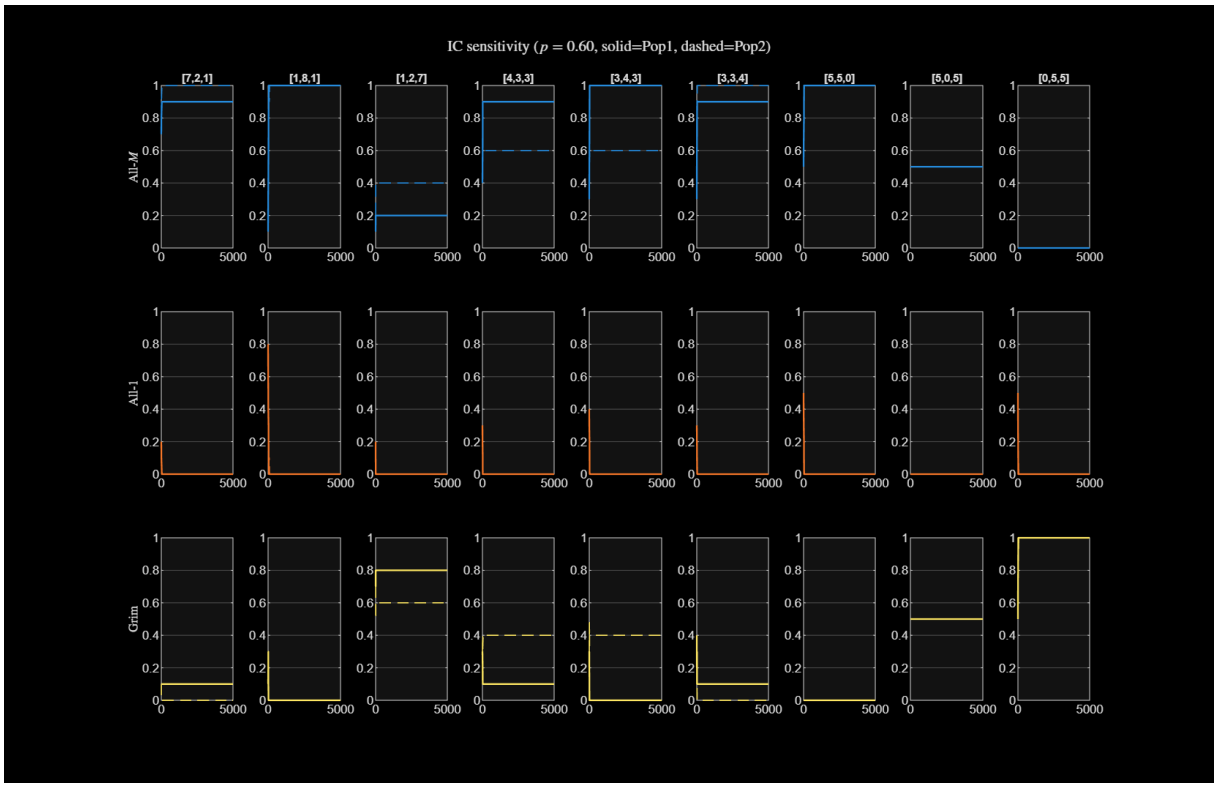


Figure 43: Initial state sensitivity ($p = 3/5$, $M = 4$, $T = 10$, $N = 10$), Variation 4.

5.6 New Phenomena vs. One-Population Model

Table 9: Comparison: one-population vs. two-population ISC (Variation 4).

Feature	One-population	Two-population
Replicator dynamics	$\dot{x}_m = x_m[(Cx)_m - x^\top Cx]$	Eqs. (4)–(5)
Accessible NE	Symmetric only	Symmetric and asymmetric
Absorbing face	1-D ($s_2 = 0$ line)	2-D ($s_2 = t_2 = 0$ face)
Total absorbing states	$N + 1$	$(N_1 + 1)(N_2 + 1)$
Stability of $\mathcal{F}$	Neutral (eigenvalue $= 0$)	Asymptotically stable
p^* for All-1 stability	$2/3$	$2/3$ (preserved)

6 Cross-Variation Remarks and Observations

6.1 The Universal $p = 2/3$ Threshold

The critical threshold $p^* = 2/3$ appears in all four variations and governs the stability of the all-defector equilibrium:

- $p > 2/3$: the All-1 equilibrium is asymptotically stable (and is an ESS where applicable). All-1 can invade and resist invasion.

- $p < 2/3$: the All-1 equilibrium is unstable. Cooperating strategies invade All-1 and drive it out. The cooperative face or cooperative absorbing states become the long-run outcome.
- $p = 2/3$: a degenerate case; All-1 is unstable to Grim-type invasion in Variations 1–3.

This threshold arises mechanically from the payoff structure: against a homogeneous All-1 population, a small Grim (or Grim-variant) mutant earns $C(\text{Grim}, 2) - C(\text{All-1}, 2) = (2 - 3p)$ more than the All-1 incumbents. The sign of this expression determines invasion success.

6.2 Absorbing State Structure Across Variations

The structure of absorbing states changes substantially across variations:

Variation	Cooperator strategy	Absorbing states
1 (All-$M/2$, numerical)	All-$M/2$	Three pure vertices only
2 (All-$M/2$, rigorous)	All-$M/2$	Three pure vertices only
3 (Five strategies)	All-M, Grim, GrimP, GrimF	$\binom{N+3}{3} + 1$ states
4 (Two populations)	All-M, Grim	$(N_1 + 1)(N_2 + 1) + 1$ states

The key insight is that whenever cooperating strategies earn *equal* payoffs against each other (as in the original All-M/Grim pair, and in the five-strategy and two-population extensions), the entire All-1-free face is absorbing. When cooperating strategies earn *unequal* payoffs in self-play (as in Variations 1–2 with All-$M/2$), only pure vertices absorb.

6.3 Grim and Cooperation

Across all variations, Grim (and strategies with similar trigger structures) is the most successful in evolutionary dynamics:

- In Variations 1–2, Grim uniquely earns TM in self-play (vs. $TM/2$ for All-$M/2$), making it the dominant absorbing vertex and the unique ESS for all p.
- In Variation 3, all four cooperating strategies earn TM against each other, but Grim best resists All-1 exploitation for $p > 2/3$.
- In Variation 4, both All-M and Grim earn TM against each other, creating a large absorbing face; neither dominates internally.

6.4 Degeneracy and Multiplicity of Equilibria

Payoff degeneracy plays a central structural role:

- In Variations 1–2, degeneracy is absent among cooperators ($C(1, 1) \neq C(3, 3)$), leading to a unique dominant strategy (Grim).
- In Variations 3–4, degeneracy ($C(i, j) = TM$ for all cooperating i, j) creates a continuum of Nash equilibria and a large absorbing face, making the evolutionary outcome highly dependent on initial conditions within the cooperative region.

6.5 One-Population vs. Two-Population Dynamics

The two-population framework of Variation 4 introduces qualitatively new phenomena:

- **Asymmetric NE:** profiles such as (Pop 1 all All-M, Pop 2 all Grim) are genuine equilibria and can be reached from generic initial conditions.
- **Stronger cooperative stability:** the cooperative face $\mathcal{F}$ is *asymptotically* stable in the transverse directions (not just neutrally stable as in the one-population case).
- **Larger absorbing set:** $(N_1 + 1)(N_2 + 1)$ cooperative absorbing states instead of $N + 1$.

Despite these differences, the $p^* = 2/3$ threshold is completely preserved.

7 Conclusions

This report has presented four complementary evolutionary analyses of the Iterated Symmetric Centipede game. The main findings, summarised across all variations, are:

1. **The $p = 2/3$ threshold is universal.** In all four variations and under both replicator dynamics and Markov chain dynamics, the critical threshold $p^* = 2/3$ governs whether the all-defector (All-1) equilibrium is stable. For $p > 2/3$, defection is an ESS; for $p < 2/3$, cooperation dominates.
2. **Grim is the strongest cooperation-sustaining strategy.** In the three-strategy settings (Variations 1–2), Grim is the unique strictly dominant ESS for all p, because it achieves the highest self-play payoff TM. In the multi-strategy setting (Variation 3), Grim variants also best resist exploitation by All-1.
3. **The All-$M/2$ modification fundamentally changes the absorbing structure** (Variations 1–2). Because All-$M/2$ earns less than Grim in self-play ($TM/2$ vs. TM), the $s_2 = 0$ face is no longer absorbing. Only the three pure vertices are. This highlights how even a moderate reduction in cooperation level can disrupt the evolutionary degeneracy structure.
4. **Adding Grim variants creates cooperative degeneracy** (Variation 3). All four cooperating strategies (All-M, Grim, Grim-Patient, Grim-Forgiver) earn identical payoffs TM against each other. This produces a continuum of Nash equilibria and a large absorbing face of dimension $\binom{N+3}{3}$, with no internal selection among cooperating strategies.
5. **Two-population dynamics access asymmetric equilibria and strengthen cooperative stability** (Variation 4). The cooperative face is asymptotically stable (not just neutrally stable) and the absorbing face is two-dimensional, reflecting independent cooperation levels of the two populations.
6. **Parameter sensitivity is consistent across variations.** Higher p promotes All-1 stability; higher M and T amplify Grim's payoff advantage; initial conditions affect convergence speed but not the qualitative long-run outcome.
7. **Learning and evolutionary interpretations are equivalent.** Under the reinforcement learning model of [6], individual agents converge to the replicator dynamic in large populations. Cooperation emerges as rational behaviour whenever $p < 2/3$.

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A Appendix: Pentagon Barycentric Coordinates (Variation 3)

The pentagon projection maps $\mathbf{x} = (x_1, \ldots, x_5) \in S_5$ to:

$$(X, Y) = \sum_{m=1}^{5} x_m \, (\cos\theta_m, \, \sin\theta_m), \quad \theta_m = \frac{\pi}{2} - \frac{2\pi(m-1)}{5}.$$

Pure state $\mathbf{e}_m$ maps to vertex $(\cos\theta_m, \sin\theta_m)$. The cooperative face $\mathcal{F} = \{x_2 = 0\}$ maps to the convex hull of $\{\theta_1, \theta_3, \theta_4, \theta_5\}$. The map is many-to-one: different $\mathbf{x}$ may have the same image. The MATLAB function `pentaBary.m` implements this projection.

B Appendix: Two-Population Replicator Dynamics — Explicit Equations (Variation 4)

Writing out Equations (4)–(5) with $\bar{\pi} = x^\top C y$:

$$
\begin{aligned}
\dot{x}_1 &= x_1\big[TM\,y_1 + 3(1-p)T\,y_2 + TM\,y_3 - \bar{\pi}\big],\\
\dot{x}_2 &= x_2\big[3pT\,y_1 + T\,y_2 + (3p+T-1)\,y_3 - \bar{\pi}\big],\\
\dot{x}_3 &= x_3\big[TM\,y_1 + (3(1-p)+T-1)\,y_2 + TM\,y_3 - \bar{\pi}\big],
\end{aligned}
$$

$$
\begin{aligned}
\dot{y}_1 &= y_1\big[TM\,x_1 + 3pT\,x_2 + TM\,x_3 - \bar{\pi}\big],\\
\dot{y}_2 &= y_2\big[3(1-p)T\,x_1 + T\,x_2 + (3(1-p)+T-1)\,x_3 - \bar{\pi}\big],\\
\dot{y}_3 &= y_3\big[TM\,x_1 + (3p+T-1)\,x_2 + TM\,x_3 - \bar{\pi}\big].
\end{aligned}
$$

Implemented in `repDyn2pop.m`.

C Appendix: Additional Strategies (Variation 3)

The five-strategy framework is easily extended:

1. **Tit-for-Tat:** begin with ρ_M; in each subsequent round play whatever OSC action the opponent used in the previous round.
2. **Generous-Tit-for-Tat:** as Tit-for-Tat but forgive defection with probability $q = (1-p)/(pM-(1-p))$.
3. **Always-m:** always play ρ_m for fixed $m \in \{1,\ldots,M\}$.